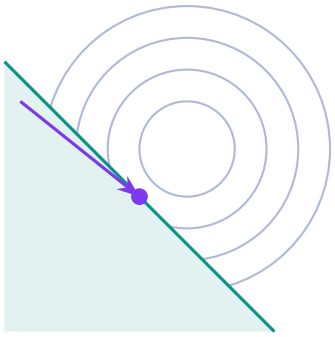
EN3330
Engineering Optimization

Study Notes - Part 2



University of Moratuwa
Semester 7

Constrained Optimization

Transformations · Lagrange multipliers · KKT and duality
Penalty and barrier methods · Feasible and infeasible-start Newton methods

Theory, original mathematical figures, detailed worked solutions,
and explanations of the reasoning behind important calculation steps.

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Notation and Study Structure

This volume is a self-contained treatment of constrained optimization. Chapters 7–9 develop the methods. Appendix A contains complete worked solutions, including the modelling exercises and numerical iterations. Each solution ends with an explanation of the important steps and the checks that justify its conclusion. Appendix B provides a compact revision reference and reproducible algorithms.

One sign convention throughout

$$\min_x f(x) \quad \text{subject to} \quad h(x) = 0, \quad g(x) \le 0, \qquad L(x, \nu, \mu) = f(x) + \nu^T h(x) + \mu^T g(x).$$

Equality multipliers ν are unrestricted; inequality multipliers μ must be nonnegative. A maximization problem is converted to minimization by negating its objective. Changing an equality from $h = 0$ to $-h = 0$ changes the sign of its multiplier, but does not change the feasible set or the optimizer.

Symbol	Meaning
n, p, m	Numbers of variables, equality constraints, and inequality constraints
J_h	Equality Jacobian; row i is ∇h_i^T
$g_f = \nabla f,\ H = \nabla^2 f$	Objective gradient and Hessian
d, α	Search direction and step length
$\delta^2 = d^T H d$	Squared Newton decrement for a feasible Newton step
ρ, t	Penalty parameter and barrier parameter, respectively
r_p, r_d	Primal-feasibility and dual-stationarity residuals

Numerical tables are recomputed from the equations using double precision. Displayed values are rounded after calculation; magnitudes below 10^{-14} are displayed as zero. Tiny negative objective gaps of the order of machine precision are round-off, not an improvement below a certified optimum.

CHAPTER 7

Transformations and Equality Constraints

Overview

Constraints restrict the directions and points that an optimizer may use. Variable transformations build some constraints directly into a parametrization. Elimination reduces the number of independent variables. Lagrange multipliers express the first-order balance between the objective gradient and the normals to equality constraints. These tools require attention to feasibility, transformation ranges, and constraint regularity.

7.1 Feasible Sets and Attainment

The general problem is

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{subject to} \quad h_i(x) = 0 \ (i = 1, \dots, p), \qquad g_j(x) \le 0 \ (j = 1, \dots, m).$$

Its feasible set is $\mathcal{F} = \{x : h(x) = 0,\ g(x) \le 0\}$. An optimal point $x^* \in \mathcal{F}$ satisfies $f(x^*) \le f(x)$ for every feasible x. A feasible set may be empty, and a finite infimum need not be attained.

The same objective with different feasible sets

For $f(x) = (x_1 - 2)^2 + (x_2 + 3)^2 + 1$, the unconstrained minimum is $(2, -3)$, with value 1. On $x_1, x_2 \ge 0$, $x_1 + x_2 \le 2$, the solution becomes $(2, 0)$, with value 10. If the lower bound on x_2 is replaced by $x_2 \le 0$, the original unconstrained minimizer is feasible and remains optimal.

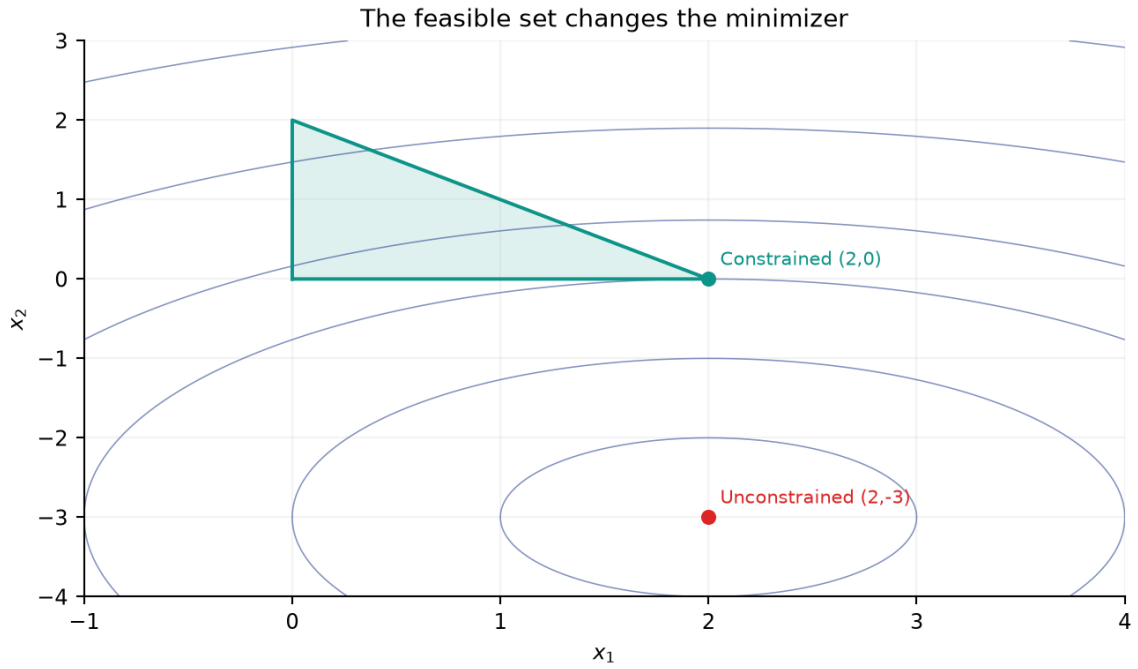


Figure 7.1: Objective contours and the triangular feasible set. Constraints can move an optimizer even when the objective is unchanged.

Check feasibility before optimizing

If $\mathcal{F}$ is nonempty and compact and f is continuous, a global minimizer exists. This does not identify the minimizer, but it justifies searching for an attained solution. Before differentiating, check whether the stated constraints can hold simultaneously.

Bounds $a \le x_i \le b$ become $a - x_i \le 0$ and $x_i - b \le 0$. An equality $h = 0$ is equivalent as a set to $h \le 0$ and $-h \le 0$, but the paired gradients are dependent. Treating it as two inequalities can therefore destroy an independence assumption used by an optimality theorem.

7.2 Bound-Preserving Transformations

Open-interval transformations

For $z \in \mathbb{R}$,

$$\sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1), \qquad \tanh z \in (-1, 1).$$

For a general interval,

$$x = T(z) = \frac{a + b}{2} + \frac{b - a}{2} \tanh z \in (a, b).$$

Then minimize $F(z) = f(T(z))$ without an explicit bound on z.

The tanh map is one-to-one onto the *open* interval. Its inverse is

$$z = \operatorname{artanh}\left(\frac{2x - a - b}{b - a}\right).$$

The derivative obeys

$$F'(z) = f'(T(z))T'(z), \qquad T'(z) = \frac{b - a}{2} \operatorname{sech}^2 z.$$

Far into either tail, $T'(z)$ is tiny. A small transformed gradient may therefore indicate saturation rather than a small gradient of the original objective.

A closed interval is not the image of a finite tanh parameter

If the constrained optimum occurs at a or b, the transformed problem may have an infimum but no finite minimizer. A solver returning a large finite z has only approximated the endpoint. Always report the original-coordinate constraint violation and optimality check.

A rational map that reaches both endpoints

$$R(z) = \frac{2z}{1+z^2} \in [-1,1], \qquad T_R(z) = \frac{a+b}{2} + \frac{b-a}{2} R(z).$$

$$R'(z) = \frac{2(1-z^2)}{(1+z^2)^2}, \qquad R(-1) = -1, \quad R(1) = 1.$$

Unlike tanh, the rational map is not globally monotone or one-to-one. The relation $R(z) = R(1/z)$ for $z \neq 0$ gives repeated representations. At $z = \pm 1$, $T'_R = 0$, so the transformed objective has a stationary point regardless of f' at the mapped endpoint. Classify it using the original problem instead of declaring every transformed stationary point optimal.

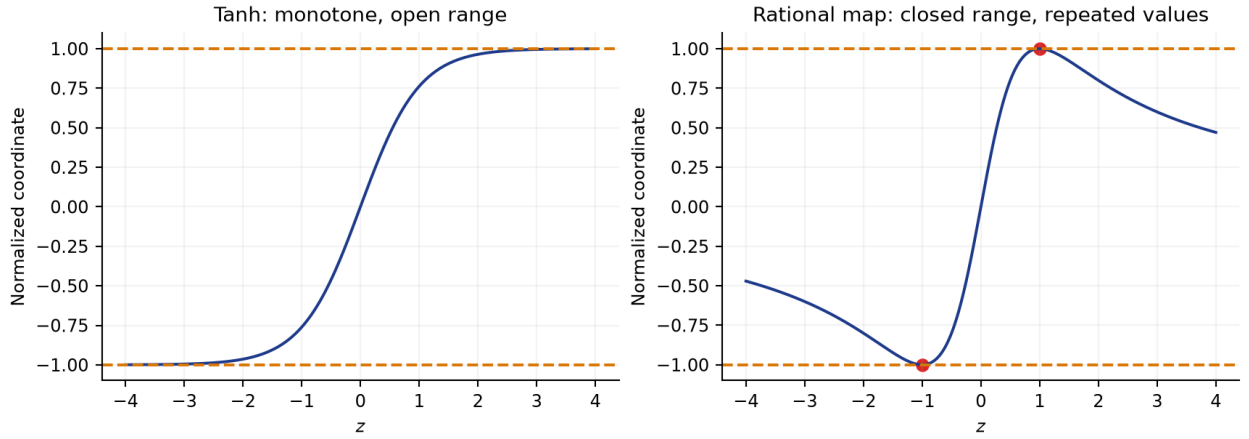


Figure 7.2: The tanh map approaches endpoints asymptotically. The rational map attains them but introduces turning points and repeated coordinates.

Map	Benefit	Limitation
Sigmoid or tanh	Smooth, monotone interior parametrization	Endpoints require infinite parameter values; tails saturate
Rational map	Both closed-interval endpoints are reached	Multiple preimages and additional stationary points
$x = d + z^2$	Enforces $x \geq d$, including the endpoint	Two-to-one except at zero; derivative vanishes at zero
$x = d - z^2$	Enforces $x \leq d$	The same derivative and multiplicity issues

For several independent bounds, transform each coordinate separately. Then $\nabla_z F = J_T^T \nabla_x f$. Nonlinear coupling constraints do not disappear merely because the individual variable bounds are enforced.

7.3 Eliminating Equality Constraints

If an equality gives $x_n = \psi(x_1, \ldots, x_{n-1})$, substitute it into the objective and every remaining constraint. The reduced search space must include the domain of ψ.

Eliminating a sphere coordinate

For $\sum_{i=1}^{n} x_i^2 = a$,

$$x_n = \pm\sqrt{a - \sum_{i=1}^{n-1} x_i^2}, \qquad \sum_{i=1}^{n-1} x_i^2 \le a.$$

Unless the objective is even in x_n, both sign branches must be considered. At a zero radicand, the branches join and derivatives of the square-root parametrization may be singular.

Elimination reduces dimension but may leave a constrained reduced problem. It can also reveal infeasibility without any numerical optimization. A complete elimination calculation must retain sign branches, radicand restrictions, and any original variable domains.

7.4 Lagrange Multipliers and Tangent Directions

For $h(x) = 0$, a differentiable feasible curve $x(s)$ through x^* has tangent $d = x'(0)$ satisfying

$$J_h(x^*)d = 0.$$

At a regular local minimum, $\nabla f(x^*)^T d = 0$ for all such tangent directions. Thus the objective gradient is normal to the feasible manifold, in the row space of J_h.

Equality-constrained first-order conditions

If the p equality gradients are linearly independent at a smooth constrained local minimizer, there exists $\nu^* \in \mathbb{R}^p$ such that

$$\nabla_x L(x^*, \nu^*) = \nabla f(x^*) + J_h(x^*)^T \nu^* = 0, \qquad h(x^*) = 0.$$

There are $n + p$ equations for $n + p$ unknowns. These equations generate candidates; they do not by themselves distinguish minima, maxima, and saddles.

For one equality in two variables, parallel gradients correspond to a shared tangent between an objective contour and the constraint curve. With several constraints, the objective gradient can be a combination of several normals. It need not be parallel to any one normal separately.

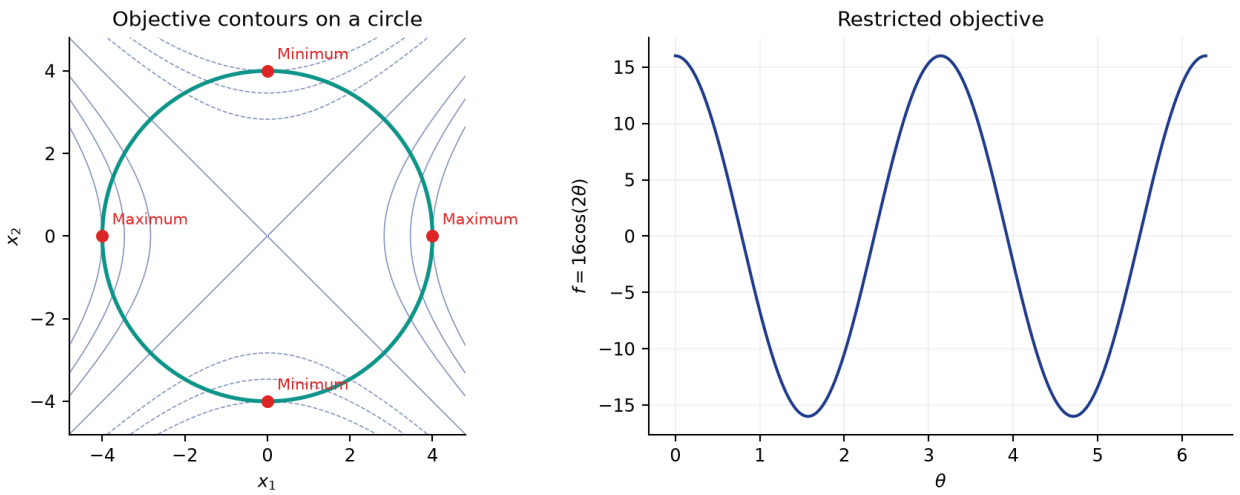


Figure 7.3: For $f = x_1^2 - x_2^2$ on $x_1^2 + x_2^2 = 16$, both maxima and minima satisfy the multiplier equations. Restricting the objective to the circle makes the distinction explicit.

7.5 Regularity and Second-Order Tests

Why a regularity condition is necessary

The problem $\min_x x$ subject to $x^2 = 0$ has the single feasible point $x = 0$. Yet $1 + \nu(2x) = 0$ cannot hold there for any finite multiplier. The constraint gradient vanishes, so the usual multiplier theorem does not apply.

Let Z have columns forming a basis for the tangent space $\ker J_h(x^*)$. At a regular candidate, use the *Lagrangian* Hessian,

$$H_L = \nabla^2 f(x^*) + \sum_i \nu_i^* \nabla^2 h_i(x^*).$$

Curvature only along feasible directions

Necessary for a local minimum: $d^T H_L d \geq 0$ for every d with $J_h d = 0$. Sufficient for a strict local minimum:

$$d^T H_L d > 0 \quad \text{for every nonzero } d \in \ker J_h, \quad \text{equivalently } Z^T H_L Z \succ 0.$$

Curvature normal to the feasible set does not determine constrained optimality. If the objective is convex and the equalities are affine, a feasible stationary point is globally optimal. Strict convexity on the feasible affine set gives uniqueness.

7.6 Interpreting Multipliers

For a constraint $a(x) = b$ written $h(x, b) = a(x) - b = 0$, let the optimal value be $v(b)$. Under differentiability and suitable stability assumptions, the envelope relation is

$$\frac{dv}{db} = -\nu^*.$$

The sign follows from $\partial L / \partial b = -\nu$, not from a memorized rule independent of convention. Multipliers depend on scaling: replacing h by ch divides its multiplier by c. State the constraint equation alongside a numerical multiplier.

Chapter Summary

Transformations enforce only the ranges they actually attain. Elimination must retain domain restrictions and all relevant branches. At a regular equality-constrained optimum, the objective gradient is balanced by constraint normals. Classify candidates using feasible-direction curvature or an independent global argument. Detailed calculations appear in Solutions A.1–A.8.

Self-check Questions

1. Why can a tanh-transformed problem fail to attain an endpoint optimum at finite z?
2. Which stationary points can a rational or square transformation introduce?
3. Why is an equality represented by two opposite inequalities problematic for LICQ?
4. Why must the Hessian of the Lagrangian be restricted to tangent directions?
5. How does rescaling a constraint change its multiplier and its sensitivity interpretation?

CHAPTER 8

KKT Conditions, Duality, and Constraint Methods

Overview

Inequality constraints may be active or inactive at the optimum. KKT conditions combine feasibility, multiplier signs, complementary slackness, and stationarity. Duality turns multipliers into lower-bound certificates. Penalty, augmented Lagrangian, and barrier methods replace a difficult constrained problem by a sequence of simpler subproblems, with different treatment of the constraint boundary.

8.1 Active Constraints and the KKT System

At a feasible point x, an inequality is *active* if $g_j(x) = 0$ and *inactive* if $g_j(x) < 0$. Equality constraints always hold at equality. An inactive inequality is locally slack, but can still limit large steps away from the point.

KKT conditions for minimization

For $L = f + \nu^T h + \mu^T g$,

$$\begin{aligned} h(x^*) = 0, \quad g(x^*) \le 0 &\qquad \text{primal feasibility,} \\ \mu^* \ge 0 &\qquad \text{dual feasibility,} \\ \mu_j^* g_j(x^*) = 0 \quad \text{for each } j &\qquad \text{complementary slackness,} \\ \nabla f(x^*) + J_h(x^*)^T \nu^* + J_g(x^*)^T \mu^* = 0 &\qquad \text{stationarity.} \end{aligned}$$

Complementary slackness means an inactive constraint has zero multiplier. An active constraint may also have zero multiplier. A geometric boundary is therefore not necessarily carrying a nonzero marginal cost.

Activity does not imply a positive multiplier

Minimize $x_1^2 + x_2^2$ with $x_1, x_2 \ge 0$ and any additional upper constraints satisfied strictly at the origin. The optimizer is $(0, 0)$. Both lower bounds are active, but all multipliers can be zero because the objective gradient is already zero.

KKT is necessary at smooth local minima under a constraint qualification. LICQ requires independent equality and active-inequality gradients. In convex problems, Slater's condition gives strong duality and, for a finite attained optimum, optimal multipliers: a point in the relative interior of the objective domain satisfies all affine constraints and all non-affine inequalities strictly.

When KKT is a global certificate

For convex f, g_j and affine h, every KKT point is globally optimal; strict convexity gives uniqueness. A constraint qualification ensures existence of multipliers at an optimum. Sufficiency of a verified convex KKT certificate needs no additional constraint qualification.

8.2 Solving by Active-Set Cases

For each plausible active set, impose $g_j = 0$ for active constraints and set $\mu_j = 0$ for inactive constraints. Solve stationarity and the equalities, then check every original inequality and every multiplier sign.

Check	Reason
Original feasibility	Solving the active equations can violate an inequality assumed inactive.
Multiplier signs	A negative multiplier contradicts the supporting-normal direction for $g \leq 0$.
All bounds	A nonnegativity boundary can contain the optimizer even if the other inequalities are slack.
Optimality, not just stationarity	Nonconvex problems can have KKT points that are not minima.
Global comparison or certificate	Compare relevant candidates and boundaries, or use convexity, duality, or a direct bound.

There can be up to 2^m active-set hypotheses, although geometry and complementary slackness often eliminate many. Numerical active-set solvers update the set intelligently rather than enumerating every possibility.

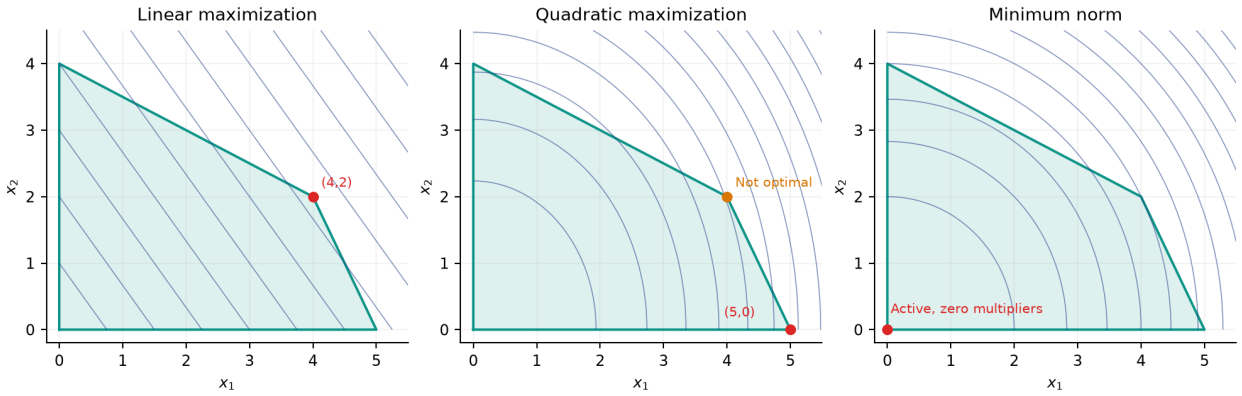


Figure 8.1: The same polygon supports different optimizers for different objectives. Maximizing a convex quadratic places the global maximum at a vertex, but not necessarily at the intersection of the two resource lines.

For a nonconvex minimization problem, second-order necessary conditions test the Lagrangian curvature on the appropriate critical directions. A simple necessary test along a smooth active edge is nonnegative curvature in both feasible tangent directions. Negative curvature along that edge rejects a candidate minimum even when first-order KKT holds.

8.3 Lagrangian Duality

The extended-valued indicator of the feasible set is zero on feasible points and $+\infty$ otherwise. For a scalar inequality,

$$\sup_{\mu \geq 0} \mu g(x) = \begin{cases} 0, & g(x) \leq 0, \\ +\infty, & g(x) > 0. \end{cases}$$

For an equality, $\sup_{\nu \in \mathbb{R}} \nu h(x)$ is zero if $h(x) = 0$, otherwise $+\infty$. Thus minimizing the worst-case Lagrangian over multipliers recovers the constrained primal problem. A supremum is the precise notation when the value is unbounded above.

Dual function and dual problem

$$q(\nu, \mu) = \inf_{x \in \operatorname{dom} f} L(x, \nu, \mu), \qquad d^* = \sup_{\nu \in \mathbb{R}^p,\ \mu \geq 0} q(\nu, \mu).$$

The inner infimum may equal $-\infty$. Multiplier values that make it unbounded below provide no useful lower bound.

For any primal-feasible x and dual-feasible (ν, μ),

$$q(\nu, \mu) \leq L(x, \nu, \mu) \leq f(x).$$

The second inequality uses $h(x) = 0$, $\mu \geq 0$, and $g(x) \leq 0$. Taking the best lower bound and best feasible objective gives *weak duality*, $d^* \leq p^*$.

Weak duality also includes zero gap

Weak duality always states a lower-bound inequality; it does not mean the gap must be positive. Strong duality means $d^* = p^*$. Equality of optimal values alone does not assert that both optima are attained. Convexity with suitable regularity often guarantees strong duality, but nonconvex problems may have either a positive or a zero gap.

The dual function is concave in its multipliers because it is an infimum of affine functions, even if the primal objective is nonconvex. A feasible primal point and a finite dual value give the computable certificate

$$0 \leq f(x) - p^* \leq f(x) - q(\nu, \mu).$$

Solution A.11 constructs an exact certificate for a circle-constrained problem.

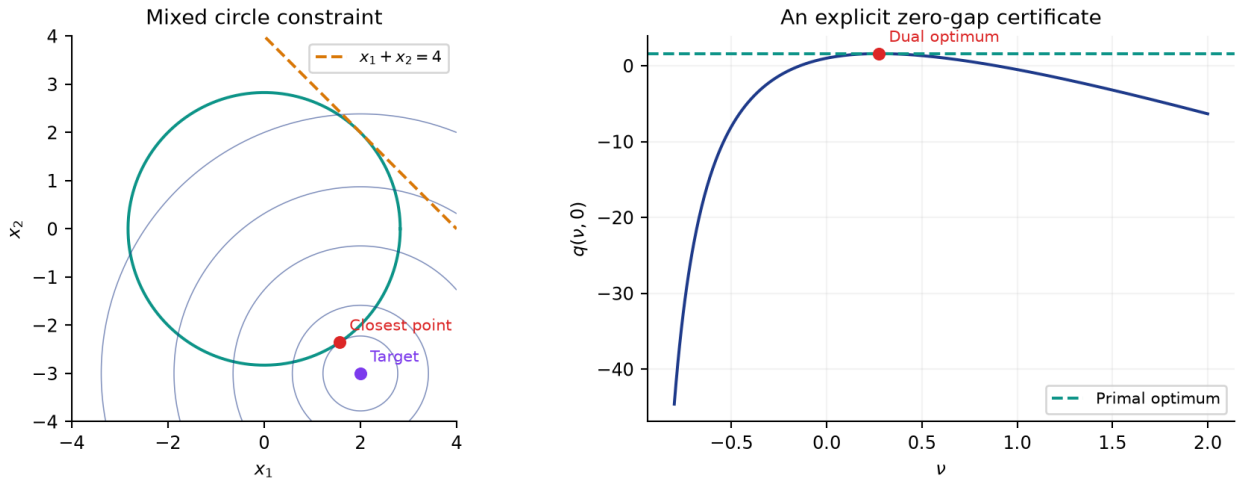


Figure 8.2: A circle-constrained projection and the dual function with inequality multiplier fixed at zero. Matching primal and dual values certify the solution directly.

8.4 Exterior Penalty Methods

Define the violation measures using the same equality and inequality convention throughout:

$$P_C(x) = \sum_i \mathbf{1}_{h_i(x) \neq 0} + \sum_j \mathbf{1}_{g_j(x) > 0},$$

$$P_Q(x) = \sum_i h_i(x)^2 + \sum_j \max\{g_j(x), 0\}^2.$$

A penalty subproblem minimizes $f(x) + \rho P(x)$. The count penalty is discontinuous and constant within any region with an unchanged violation pattern. It gives no smooth gradient pointing toward feasibility. Squared violation supplies a magnitude-dependent force.

Penalty	Advantage	Limitation
Count	A fixed jump can distinguish feasible and infeasible designs	Discontinuous; no violation-distance gradient; finite exactness needs additional assumptions
Quadratic	Differentiable guidance when constraints are smooth	Usually approaches feasibility only as $\rho \to \infty$; can become ill-conditioned
Mixed count and quadratic	Retains the jump and adds exterior guidance	The jump remains discontinuous; a smooth solver is not automatically suitable

The scalar function $(\max\{u, 0\})^2$ is continuously differentiable but not twice differentiable at zero. A smooth-looking penalty objective need not have a continuous Hessian across an inequality boundary.

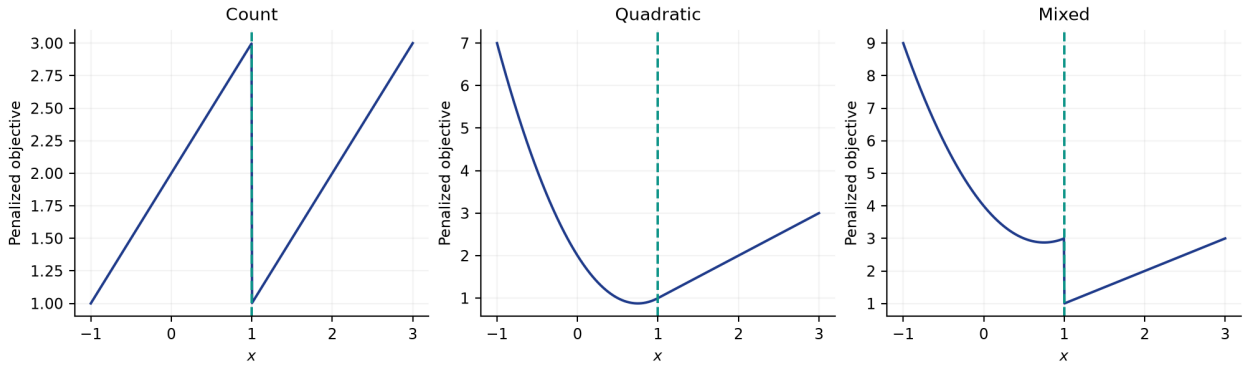


Figure 8.3: Count, quadratic, and mixed penalties for minimizing x subject to $x \geq 1$, with weight 2. The count jump does not control an objective that decreases without bound far into the infeasible region.

Penalty continuation

Choose $\rho_0 > 0$ and $\gamma > 1$. Approximately solve

$$x_{k+1} \approx \arg\min_x \{f(x) + \rho_k P_Q(x)\}, \qquad \rho_{k+1} = \gamma \rho_k.$$

Warm-start each subproblem from the preceding solution. Track constraint violation and inner-solver optimality, not just the change in x.

For an affine equality $Ax = b$, the Hessian contribution of $(\rho/2)\|Ax - b\|^2$ is $\rho A^T A$. It grows normal to the feasible set but not along its tangent directions, often creating a poorly conditioned subproblem. Convergence to a constrained solution requires assumptions and sufficiently accurate subproblem solves; increasing ρ alone is not a universal guarantee.

8.5 Augmented Lagrangian Methods

For equality constraints, combine a multiplier term and a quadratic penalty:

Augmented Lagrangian and multiplier update

$$\mathcal{L}_\rho(x,\nu) = f(x) + \nu^T h(x) + \frac{\rho}{2}\|h(x)\|_2^2,$$
$$x_{k+1} \approx \arg\min_x \mathcal{L}_{\rho_k}(x,\nu_k), \qquad \nu_{k+1} = \nu_k + \rho_k h(x_{k+1}).$$

At an accurately solved subproblem,

$$\nabla f(x_{k+1}) + J_h(x_{k+1})^T\big(\nu_k + \rho_k h(x_{k+1})\big) \approx 0.$$

The quantity in parentheses is precisely the updated multiplier. This is why the update improves the stationarity balance rather than merely imposing a larger penalty. A suitable multiplier can remove the bias of a finite quadratic penalty.

Moderate penalty does not mean finite-step exactness

The method can converge with bounded ρ under appropriate assumptions. It generally learns the correct multiplier iteratively, rather than producing exact feasibility at the first finite iteration. Increase ρ when feasibility is not improving sufficiently; making it unnecessarily large can still harm conditioning.

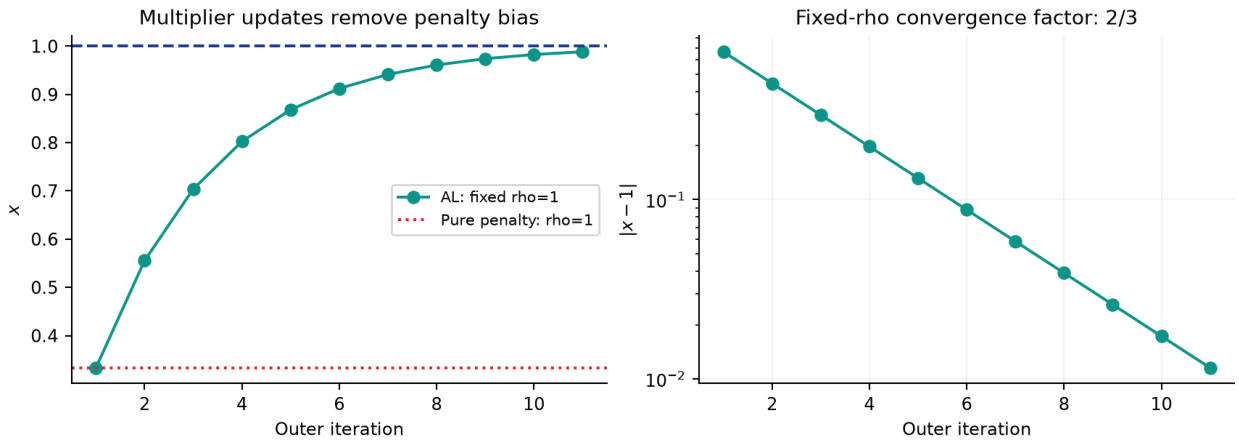


Figure 8.4: For $\min x^2$ subject to $x = 1$, a fixed quadratic penalty has bias, while multiplier updates converge with $\rho = 1$. The complete scalar calculation is in Solution A.14.

The formula above is for equalities. For $g \le 0$, one valid extension introduces slacks $s \ge 0$ and equalities $g(x) + s = 0$, then handles the slack bounds explicitly. Simply applying an unrestricted equality multiplier update to an inequality is not generally valid.

8.6 Interior Barriers and the Central Path

For strictly feasible inequalities $g_j(x) < 0$, define

$$\phi_{\log}(x) = -\sum_j \log(-g_j(x)), \qquad \phi_{\text{inv}}(x) = -\sum_j \frac{1}{g_j(x)}.$$

Both diverge as an inequality boundary is approached from the interior. The inverse barrier is positive, but the log barrier can be negative when slacks exceed one. Nonnegativity is not a required property of a barrier.

Equivalent barrier formulations

$$\min_x f(x) + \frac{1}{t}\phi(x) \quad \text{subject to } Ax = b \quad \Longleftrightarrow \quad \min_x tf(x) + \phi(x) \quad \text{subject to } Ax = b, \quad t > 0.$$

The minimizers for varying t form the central path. Increasing t weakens the barrier relative to the original objective.

The method must preserve strict inequality feasibility, not merely $g \leq 0$. If affine equalities are retained, the subproblem is still equality constrained. It becomes fully unconstrained only after an appropriate equality elimination or parametrization.

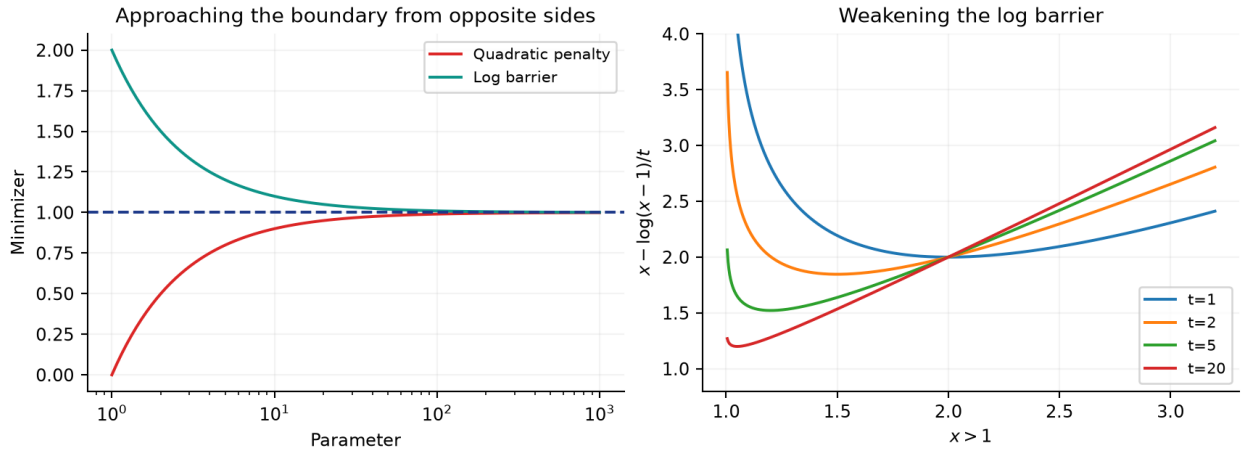


Figure 8.5: For $\min x$ subject to $x \geq 1$, the quadratic penalty minimizer approaches from below and the log-barrier minimizer from above.

8.6.1 Perturbed KKT conditions and a gap bound

At an exact minimizer of $tf + \phi$ with equality multiplier w,

$$t\nabla f - \sum_j \frac{1}{g_j}\nabla g_j + A^T w = 0.$$

Set $\mu_j = -1/(tg_j) > 0$ and $\nu = w/t$. Dividing by t yields the ordinary stationarity equation, but complementary slackness becomes

$$\mu_j g_j = -\frac{1}{t} \quad \text{or} \quad \mu_j s_j = \frac{1}{t}, \qquad s_j = -g_j > 0.$$

For a convex problem, the stationarity point minimizes the Lagrangian. Therefore

$$q(\nu, \mu) = L(x, \nu, \mu) = f(x) - \frac{m}{t}, \qquad 0 \leq f(x) - p^* \leq \frac{m}{t}.$$

The bound assumes an accurately centered convex subproblem

The bound m/t is not a certificate for any arbitrary strictly feasible point. A numerical implementation must also control the inner centering error or construct an explicit dual lower bound. Small successive steps alone do not establish a small optimization gap.

8.6.2 Newton centering with affine inequalities

Write $g(x) = Ax - b$, with positive slack $s = b - Ax$. Then

$$\nabla\phi = A^T s^{-1}, \qquad \nabla^2\phi = A^T \operatorname{diag}(s_j^{-2})A,$$

where inverse and square are componentwise. For a direction d, only rows with $a_j^T d > 0$ limit a positive step:

$$\alpha_{\max} = \min_{a_j^T d>0} \frac{s_j}{a_j^T d}.$$

Choose a strict fraction, such as $0.99\alpha_{\max}$, when this bound is finite, then use objective backtracking. This preserves the log domain before testing decrease.

8.7 The Barrier Algorithm in Practice

1. Find a point in the objective domain with $Ax = b$ and strictly positive inequality slacks. A Phase I problem can maximize a common slack margin. A penalty solver reaching a boundary point does not automatically supply a strict interior point.
2. For the current t, approximately minimize $tf + \phi$, using domain-preserving Newton steps.
3. Check inner residuals and the appropriate gap certificate. In the ideal convex centered case, stop when $m/t \le \varepsilon$.
4. Otherwise set $t \leftarrow \gamma t$, with $\gamma > 1$, and warm-start the next centering solve.

Small γ gives many easy outer steps; very large γ gives fewer, harder centering problems. A useful implementation balances total Newton work and numerical stability.

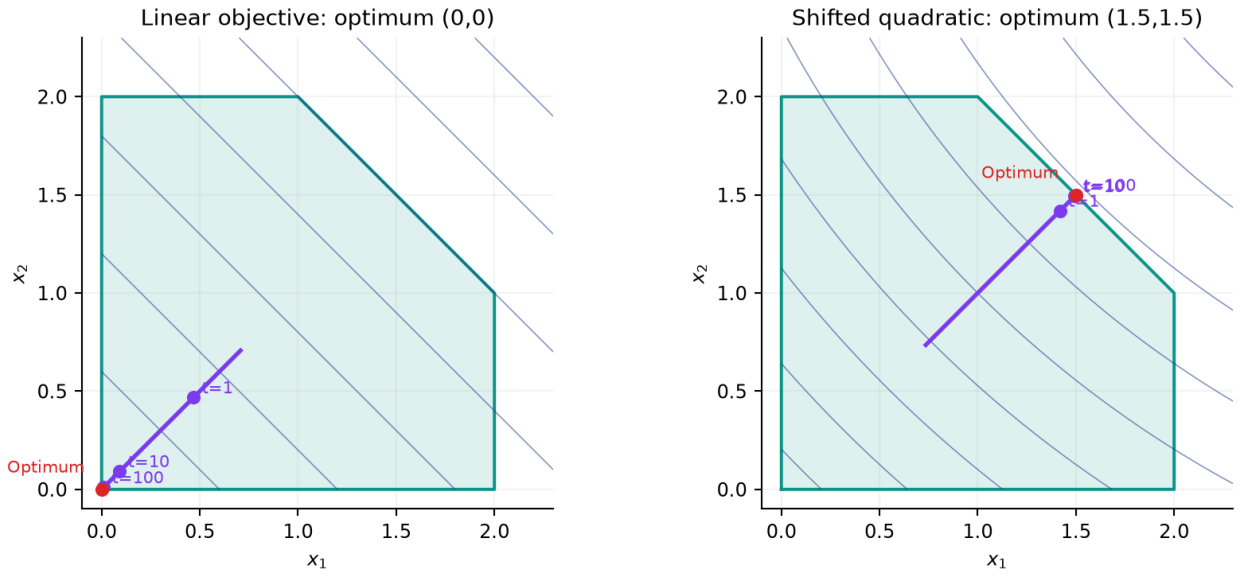


Figure 8.6: Central paths for two objectives on $0 \le x_i \le 2$, $x_1 + x_2 \le 3$. The linear objective approaches $(0, 0)$; the shifted quadratic approaches $(1.5, 1.5)$. Symmetry makes both paths lie on the diagonal.

8.8 Choosing a Constraint Method

Property	Quadratic penalty	Augmented Lagrangian	Log barrier
Initial inequality feasibility	Not required	Depends on formulation	Strictly required
Typical iterates	May be infeasible	May be infeasible	Strictly feasible inequalities
Equality constraints	Penalized	Multiplier plus penalty	Retained or eliminated
Main parameter	Large ρ reduces violation	Multiplier update can allow moderate ρ	Large t weakens barrier
Main risk	Ill-conditioning	Inaccurate inner solves or unsuitable updates	Loss of domain, difficult centering
Stopping evidence	Violation and stationarity	Violation and KKT residual	Centering accuracy and primal-dual gap

Chapter Summary

Active-set algebra must be followed by feasibility, sign, and optimality checks. KKT is sufficient globally in the convex setting, but not generally. Dual values certify lower bounds. Penalties allow violations, augmented multipliers learn constraint forces, and barriers preserve strict interior feasibility while approaching boundary optima. Solutions A.9–A.15 work through the calculations in detail.

Self-check Questions

1. Can an active inequality have zero multiplier? Can an inactive one have a positive multiplier?
2. Why is maximizing a convex quadratic not a convex optimization problem?
3. Prove weak duality using the sign of $\mu^T g(x)$.
4. Why is a count penalty not smooth, and why can finite exactness fail for an unbounded objective?
5. Explain why a log barrier need not be nonnegative.
6. Derive $\mu_j s_j = 1/t$ and list the assumptions behind the bound m/t.

CHAPTER 9

Newton Methods for Constrained Problems

Overview

Newton's method can be understood as solving a local quadratic model or applying root finding to stationarity equations. With affine equality constraints, the Newton step solves a KKT linear system. A feasible start gives tangent steps and objective descent. An infeasible start instead reduces a combined primal-dual residual, so the objective may initially increase while feasibility improves.

9.1 From Root Finding to a Quadratic Model

For a scalar root $F(x) = 0$, linearize $F(x + d) \approx F(x) + F'(x)d$. The Newton update is $d = -F/F'$. Optimization uses $F = f'$, giving $d = -f'/f''$, not $-f/f'$.

Multivariate Newton model

At x, write $g_f = \nabla f(x)$, $H = \nabla^2 f(x)$, and

$$q(d) = f(x) + g_f^T d + \frac{1}{2} d^T H d.$$

Its stationary equation is $Hd = -g_f$. If $H \succ 0$, this is the unique model minimizer and $g_f^T d < 0$ whenever $g_f \neq 0$.

Solve the linear system rather than explicitly computing H^{-1}. A negative or indefinite Hessian can turn the model stationary point into a maximum or saddle; merely solving $Hd = -g_f$ does not establish a minimization step.

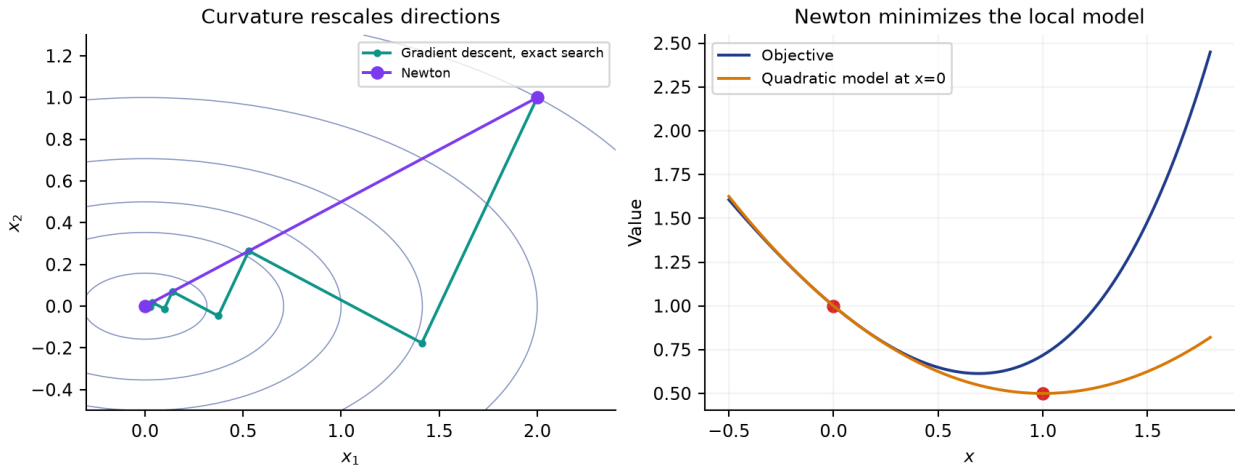


Figure 9.1: Newton rescales each direction by curvature. On a positive-definite quadratic the model is exact, so a full Newton step reaches the minimizer in one iteration.

9.2 Newton Decrement and Damping

For an unconstrained Newton direction with positive-definite Hessian,

$$\delta^2 = d^T H d = -g_f^T d = g_f^T H^{-1} g_f.$$

The predicted model decrease is

$$q(0) - q(d) = -g_f^T d - \frac{1}{2} d^T H d = \frac{1}{2}\delta^2.$$

This is an exact statement about the quadratic model. It is an exact objective gap for a positive-definite quadratic, and often a useful local gap estimate for a smooth nonlinear objective. It is not an unconditional global error bound for an arbitrary convex function.

Damped Newton with Armijo backtracking

Choose $c \in (0, 1/2)$, $\beta \in (0, 1)$. Start $\alpha = 1$ and reduce $\alpha \leftarrow \beta\alpha$ until the trial point is in the domain and

$$f(x + \alpha d) \le f(x) + c\alpha g_f^T d = f(x) - c\alpha\delta^2.$$

Then update $x \leftarrow x + \alpha d$.

Requiring a fraction of the predicted directional decrease prevents acceptance of arbitrarily unproductive steps simply because f decreased by a tiny amount. Domain checks and descent-direction checks must precede the line search. If H is not positive definite, damping alone does not necessarily fix the direction; modify the model or use a trust region.

9.3 Failure Modes and Convergence Assumptions

Objective	Newton behaviour	Reason
$x^4/4$	$x_{k+1} = 2x_k/3$, only linear convergence	Curvature vanishes at the minimizer
$x^3 - 3x$	Some starts converge to the maximum at -1	Newton solves stationarity, without classifying it
$\sqrt{1+x^2}$	$x_{k+1} = -x_k^3$, divergent for $\lvert x_0\rvert > 1$	A full local-model step can be poor far from the solution

Local quadratic convergence typically requires a nonsingular positive-definite Hessian at the solution, sufficient Hessian regularity such as local Lipschitz continuity, and a sufficiently close starting point. Twice differentiability alone does not guarantee global convergence or a quadratic local rate.

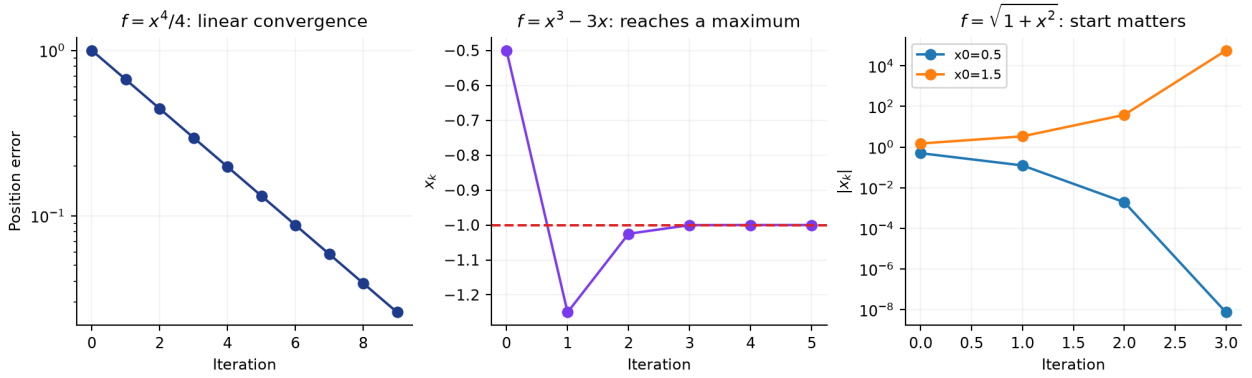


Figure 9.2: Three distinct issues: degenerate curvature, convergence to the wrong stationary point, and divergence caused by a distant start.

9.4 Feasible Newton Steps with Affine Equalities

Consider min $f(x)$ subject to $Ax = b$, with $A \in \mathbb{R}^{p\times n}$ of full row rank. At a feasible point, the quadratic subproblem is

$$\min_d g_f^T d + \frac{1}{2} d^T H d \quad \text{subject to } Ad = 0.$$

Its Lagrangian is $g_f^T d + \frac{1}{2} d^T H d + \nu_+^T A d$. Differentiating with respect to d and the multiplier gives

Feasible-start KKT system

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} d \\ \nu_+ \end{bmatrix} = \begin{bmatrix} -g_f \\ 0 \end{bmatrix}.$$

The subscript $+$ indicates the multiplier associated with the new quadratic model, not a multiplier increment.

The last row ensures $A(x + \alpha d) = b$ for every α. Premultiplying stationarity by d^T gives

$$g_f^T d + d^T H d + (Ad)^T \nu_+ = 0 \quad \Longrightarrow \quad g_f^T d = -d^T H d.$$

Thus a nonzero step is a descent direction if the Hessian is positive definite on the null space of A. The constrained decrement is again $\delta^2 = d^T H d$.

9.4.1 Why reduced curvature controls the KKT matrix

Let Z span ker A. If $Z^T H Z \succ 0$ and A has full row rank, the KKT matrix is nonsingular. To prove it, suppose

$$Hu + A^T v = 0, \qquad Au = 0.$$

Taking the inner product with u yields $u^T H u = 0$. Positive tangent curvature forces $u = 0$. Then $A^T v = 0$, and full row rank forces $v = 0$. Convexity alone gives a semidefinite Hessian and is not enough to establish this property.

A singular full Hessian can still be acceptable

For $H = \text{diag}(0, 2)$ and $A = [1\ 0]$, the feasible directions are $d_1 = 0$. Curvature along the remaining direction is strictly positive, so the KKT matrix is nonsingular despite the singular H.

9.4.2 Feasible-start algorithm

Start with $Ax_0 = b$ and $x_0 \in \text{dom } f$. At each iteration, solve the KKT system, evaluate $\delta^2/2$, backtrack on f, and update x. Monitor both the decrement and KKT residuals. For a quadratic objective with positive tangent curvature, a full step solves the problem exactly in exact arithmetic.

9.5 Infeasible-Start Newton as KKT Root Finding

When $Ax \neq b$, objective descent alone is not an appropriate measure of progress. The optimality equations are

$$r_d(x, \nu) = \nabla f(x) + A^T \nu = 0, \qquad r_p(x) = Ax - b = 0.$$

Stack $r = (r_d, r_p)$. Its Jacobian is the KKT matrix, so linearizing r gives

Primal-dual increment system

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \nu \end{bmatrix} = - \begin{bmatrix} r_d \\ r_p \end{bmatrix}.$$

Update both variables using the same accepted step: $x_+ = x + \alpha \Delta x$, $\nu_+ = \nu + \alpha \Delta \nu$.

The equivalent direct-multiplier system is

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \widehat{\nu}_+ \end{bmatrix} = - \begin{bmatrix} \nabla f \\ r_p \end{bmatrix}, \qquad \Delta \nu = \widehat{\nu}_+ - \nu.$$

Do not mix the two multiplier meanings

The first system returns $\Delta \nu$; the second returns $\widehat{\nu}_+$. If the second system is used, adding its multiplier directly to the current ν applies the wrong update. With a damped step, interpolate using $\Delta \nu = \widehat{\nu}_+ - \nu$.

9.5.1 Residual backtracking and feasibility restoration

Choose $c \in (0, 1)$, $\beta \in (0, 1)$, and reduce a trial $\alpha = 1$ until the domain test passes and

$$\|r(x + \alpha \Delta x, \nu + \alpha \Delta \nu)\|_2 \leq (1 - c\alpha) \|r(x, \nu)\|_2.$$

The linear model predicts $r_+ \approx (1 - \alpha) r$, motivating this acceptance rule. The primal part has an exact relation:

$$r_p(x + \alpha \Delta x) = r_p(x) + \alpha A \Delta x = (1 - \alpha) r_p(x).$$

A full step therefore restores affine feasibility exactly in exact arithmetic. A damped step reduces violation by a factor of $1 - \alpha$. Neither statement says that the nonlinear dual residual vanishes after one step.

The objective may increase while the method succeeds

For $\min x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$, the infeasible point $(0, 0)$ has objective zero. Every feasible point has positive objective. Moving to the true optimum $(1/2, 1/2)$ increases the objective to $1/2$, but removes infeasibility and satisfies stationarity. Compare residuals, not the infeasible starting objective, when judging progress.

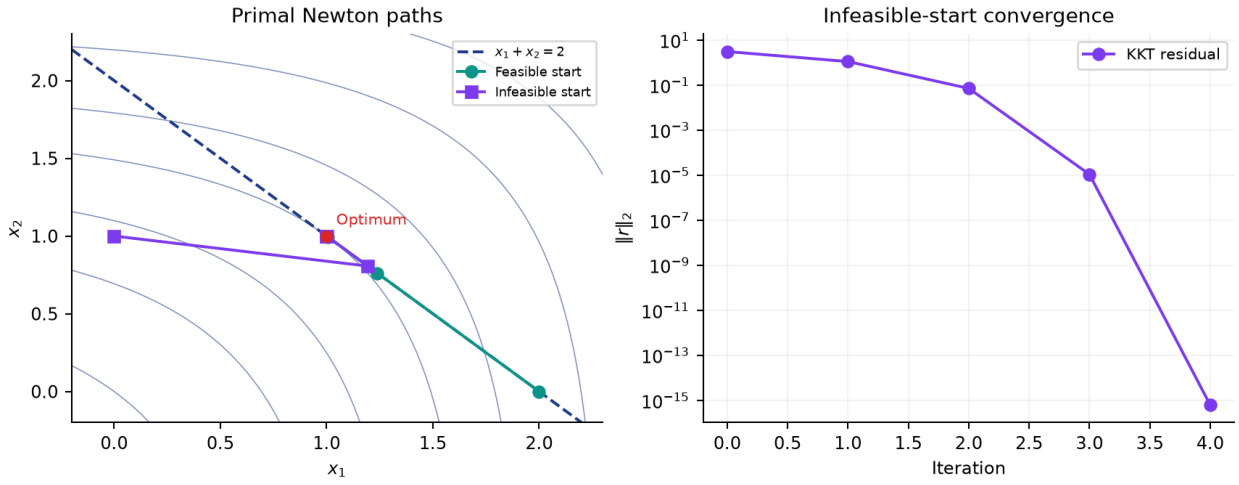


Figure 9.3: Feasible and infeasible-start paths for $e^{x_1} + e^{x_2}$ with $x_1 + x_2 = 2$. The residual graph measures simultaneous feasibility and stationarity.

In floating-point code, stop using tolerances on $\|r_p\|$ and $\|r_d\|$, rather than exact equality tests. For differently scaled physical units, use scaled residuals and compatible tolerances.

9.6 Connection to Sequential Quadratic Programming

For nonlinear equalities $h(x) = 0$, the first-order constraint model is

$$h(x + d) \approx h(x) + J_h(x)d = 0.$$

SQP combines it with a quadratic objective model, ideally using the Lagrangian Hessian or a suitable positive-definite approximation. Affine constraints make the linearization exact; nonlinear ones generally do not remain exactly feasible after a finite step. Globalization can use a merit function, a filter, or trust-region rules. Inequalities require corresponding active-set or interior-point machinery in the QP subproblem.

9.7 Engineering Interpretation: Economic Dispatch

For generator costs $C_i(P_i)$ and demand D,

$$\min_P \sum_i C_i(P_i) \quad \text{subject to} \quad \sum_i P_i = D.$$

Stationarity is $C_i'(P_i) + \nu = 0$. All generators not limited by inequality bounds therefore have the same marginal cost. The system marginal price is $-\nu$, consistent with the sensitivity $dC^*/dD = -\nu$. A unit at a capacity limit requires an additional nonnegative bound multiplier and need not have the same individual marginal cost.

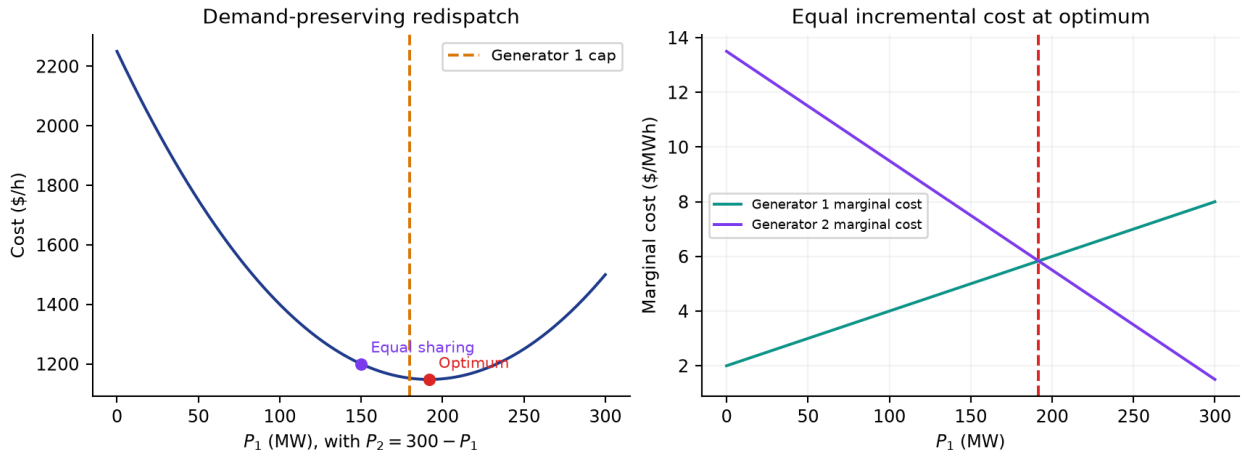


Figure 9.4: Redispatch preserves total demand while moving generation from the initially more expensive marginal source to the cheaper one. The optimum equalizes marginal costs unless a capacity bound is active.

Chapter Summary

Newton solves a local model, so curvature and globalization matter. Feasible equality Newton solves a tangent-space QP and backtracks on the objective. Infeasible-start Newton solves the primal-dual KKT equations and backtracks on residual norm. Distinguish multiplier increments from new multiplier estimates. Solutions A.16–A.22 include detailed calculations, iteration tables, and verification.

Self-check Questions

1. Derive $Hd = -\nabla f$ from the quadratic model and explain the positive-definiteness requirement.
2. Why is $\delta^2/2$ a model decrease rather than an automatic global error bound?
3. Prove that feasible-start equality Newton preserves $Ax = b$ for any step length.
4. Give the condition on $Z^T H Z$ that makes the KKT matrix nonsingular.
5. Derive $r_{p,+} = (1 - \alpha) r_p$ for infeasible-start Newton.
6. Why can an objective-increasing step be accepted by the infeasible-start method?

Appendix A

Detailed Worked Solutions

Every calculation starts from a complete problem statement. Results are followed by feasibility and optimality checks. The final green explanation in each solution states why the important algebraic or modelling steps are justified.

Solutions	Main reasoning skill
A.1–A.3	Transformation ranges, endpoint attainment, and feasibility before optimization
A.4–A.8	Multiplier equations, branch classification, and modelling inequalities correctly
A.9–A.12	Full KKT checks, active sets, convex sufficiency, and dual certificates
A.13–A.15	Parameter limits, multiplier updates, and central-path calculations
A.16–A.17	Model curvature, step acceptance, and numerical convergence
A.18–A.20	KKT linear algebra, feasible directions, and sensitivity
A.21–A.22	Coupled primal-dual steps and residual-based progress

A reliable calculation sequence

State domains and signs. Differentiate the correct function. Solve the candidate equations. Check all constraints and multiplier signs. Classify local behaviour or provide a global certificate. Interpret the answer in the original variables and units.

A.1 Bounded scalar optimization by transformation

Problem

Minimize $f(x) = x^2 \cos x$ for $2 \leq x \leq 7$. Formulate both a tanh transformation and a rational transformation, compute the solution, and check it in the original interval.

1. Establish the correct coordinate maps.

The midpoint is $(2 + 7)/2 = 4.5$, and the half-width is $(7 - 2)/2 = 2.5$. Thus

$$T(z) = 4.5 + 2.5 \tanh z, \qquad T_R(z) = 4.5 + \frac{5z}{1 + z^2}.$$

The transformed objectives are $F(z) = T(z)^2 \cos(T(z))$ and $F_R(z) = T_R(z)^2 \cos(T_R(z))$. The midpoint must be 4.5 for the image to have the intended endpoints 2 and 7.

2. Solve the original stationary equation and include endpoints.

$$f'(x) = 2x \cos x - x^2 \sin x = x(2 \cos x - x \sin x).$$

Since $x \geq 2$, the factor x is not zero. Bracketed root solves of $2 \cos x - x \sin x = 0$ give the two stationary points in the interval. They lie in $(\pi, 3\pi/2)$ and $(2\pi, 7)$; the equation $\tan x = 2/x$, where defined, has at most one root in each of these intervals because its left side increases and its right side decreases.

Candidate	x	$f(x)$
Left endpoint	2	-1.664587
Stationary point	3.643597	-11.637829
Stationary point	6.578334	41.403242
Right endpoint	7	36.94121

The global minimizer is $x^* = 3.6435971674$, with $f(x^*) = -11.6378292118$.

3. Recover the transformed coordinates.

$$z^* = \operatorname{artanh}\left(\frac{x^* - 4.5}{2.5}\right) = -0.3569912902.$$

For the rational map, set $y = (x^* - 4.5)/2.5$. Solving $y = 2z/(1 + z^2)$ gives

$$yz^2 - 2z + y = 0, \qquad z = \frac{1 \pm \sqrt{1 - y^2}}{y}.$$

The two representations are approximately -0.1766238313 and -5.6617501323. Substitution into the maps recovers the same original minimizer.

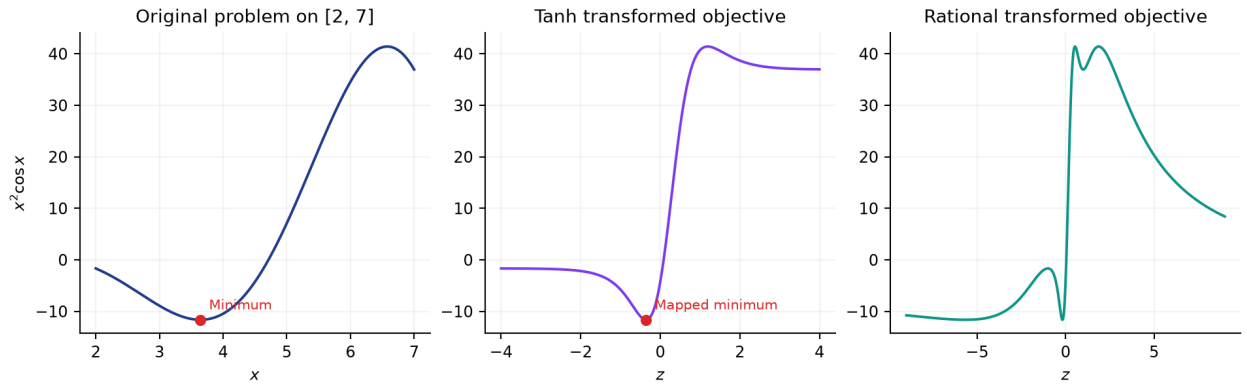


Figure A.1: The original objective and the two transformed versions. Repeated coordinates in the rational map produce repeated copies of the same original minimum.

Why the Steps Work

The chain rule explains why interior stationary points correspond under tanh: $T' \neq 0$ at finite z. The rational map can add stationary points through $T'_R = 0$, so checking only its derivative is insufficient. Comparing all original stationary points and both endpoints is a global check on a continuous function over a compact interval. Mapping back is essential because the requested design variable is x, not z.

A.2 A tanh-transformed problem whose optimum is an endpoint

Problem

Minimize $f = x_1^2 + x_2^2 + 1$ subject to $2 \le x_1 \le 4$ and $3 \le x_2 \le 5$. Transform the bounds and determine whether the transformed minimum is attained.

The coordinate maps are $x_1 = 3 + \tanh z_1$, $x_2 = 4 + \tanh z_2$. Hence

$$F(z_1, z_2) = (3 + \tanh z_1)^2 + (4 + \tanh z_2)^2 + 1.$$

For every finite z, $x_1 > 2$ and $x_2 > 3$, and

$$\frac{\partial F}{\partial z_1} = 2(3 + \tanh z_1)\,\operatorname{sech}^2 z_1 > 0, \qquad \frac{\partial F}{\partial z_2} = 2(4 + \tanh z_2)\,\operatorname{sech}^2 z_2 > 0.$$

Both coordinates must decrease. In the original closed box, $x^* = (2, 3)$, with $f^* = 14$. In transformed coordinates,

$$F(z) \longrightarrow 14 \quad \text{as } z_1, z_2 \longrightarrow -\infty,$$

but $F(z) > 14$ for every finite pair. There is no finite transformed minimizer.

Why the Steps Work

The original objective increases with each positive coordinate, so the lower-left corner is optimal. A tanh map covers only the open box, excluding exactly that corner. Small transformed derivatives near negative infinity indicate saturation. They do not prove that a finite transformed point is the exact original solution.

A.3 Variable elimination and an infeasible pair of equalities

Problem

Consider $f = x_1^2 + x_1x_2^2 + x_3^2 + 1$, subject to

$$x_1^2 + x_2^2 + x_3^2 = 16, \qquad 5x_1^2 + 4x_2^4 + x_3^2 = 9.$$

Eliminate variables and decide whether a real solution exists.

1. Eliminate the common squared coordinate.

From the second equality,

$$x_3^2 = 9 - 5x_1^2 - 4x_2^4 \geq 0.$$

Substitution into the first equality gives

$$-4x_1^2 + x_2^2 - 4x_2^4 - 7 = 0.$$

Equivalently,

$$4x_1^2 + 4x_2^4 - x_2^2 + 7 = 0.$$

2. Complete the square instead of optimizing an imaginary branch.

Let $s = x_2^2 \geq 0$. Then

$$4x_1^2 + 4\left(s - \frac{1}{8}\right)^2 + \frac{111}{16} = 0.$$

Every term on the left is nonnegative, and the constant is strictly positive. The equation is impossible. The feasible set is empty, so this problem has no real optimizer.

3. Contrast with the problem containing only the sphere equality.

If the second equality is absent, eliminate $x_3^2 = 16 - x_1^2 - x_2^2$, retaining $x_1^2 + x_2^2 \leq 16$. The reduced objective is

$$f = 17 + (x_1 - 1)x_2^2.$$

For $x_1 < 1$, its minimum at fixed x_1 uses the largest admissible x_2^2, namely $16 - x_1^2$, giving $x_3 = 0$. For $x_1 \geq 1$, choose $x_2 = 0$, giving 17. In the first region,

$$F(x_1) = 17 + (x_1 - 1)(16 - x_1^2), \quad F' = 16 - 3x_1^2 + 2x_1.$$

The minimizing root is $x_1 = -2$; the other root $8/3$ lies outside $x_1 < 1$. Checking endpoints gives

$$x^* = (-2,\ \pm 2\sqrt{3},\ 0), \qquad f^* = -19.$$

Why the Steps Work

Elimination must preserve nonnegative radicands and all original equalities. A negative square-root radicand is evidence of infeasibility, not a candidate to continue differentiating. In the one-equality variant, choosing the largest possible x_2^2 is justified only when its coefficient $x_1 - 1$ is negative. The sign split is what makes the global reduction valid.

A.4 Classifying multiplier candidates on a circle

Problem

Find the constrained extrema of $f = x_1^2 - x_2^2$ subject to $x_1^2 + x_2^2 = 16$.

Use $L = x_1^2 - x_2^2 + \nu(x_1^2 + x_2^2 - 16)$. Stationarity gives

$$2(1+\nu)x_1 = 0, \qquad 2(\nu-1)x_2 = 0.$$

If $x_1 \neq 0$, then $\nu = -1$; the second equation forces $x_2 = 0$. Feasibility gives $x_1 = \pm 4$, with value 16. If $x_2 \neq 0$, then $\nu = 1$, forcing $x_1 = 0$, $x_2 = \pm 4$, with value -16. Both coordinates cannot be zero on the circle, nor can both be nonzero because that would require $\nu = -1$ and $\nu = 1$ simultaneously.

Points	ν	Classification
$(\pm 4, 0)$	-1	Global maxima
$(0, \pm 4)$	1	Global minima

The global check is immediate from $f = 16 - 2x_2^2$ and $0 \leq x_2^2 \leq 16$. For a second-order check, at $(0, 4)$ the tangent directions have $d_2 = 0$, and $\nabla^2 L = \mathrm{diag}(4, 0)$, giving strictly positive tangent curvature.

Why the Steps Work

The product equations require a case split; dividing by a coordinate would discard valid solutions with that coordinate zero. Lagrange equations locate both maxima and minima. Substitution of the constraint into the objective produces an exact global range, making the final classification stronger than a local stationary-point test.

A.5 Minimum norm with a prescribed coordinate sum

Problem

Minimize $x_1^2 + x_2^2$ subject to $x_1 + x_2 = 5$.

$$L = x_1^2 + x_2^2 + \nu(x_1 + x_2 - 5).$$

Stationarity and feasibility are

$$2x_1 + \nu = 0, \qquad 2x_2 + \nu = 0, \qquad x_1 + x_2 = 5.$$

Subtracting the first two equations eliminates the common multiplier: $x_1 = x_2$. Thus

$$x_1^* = x_2^* = \frac{5}{2}, \qquad \nu^* = -5, \qquad f^* = \frac{25}{2}.$$

Elimination gives the independent check

$$x_1^2 + (5 - x_1)^2 = 2\left(x_1 - \frac{5}{2}\right)^2 + \frac{25}{2}.$$

The square is nonnegative and vanishes only at $x_1 = 5/2$, proving the unique global minimum. More generally, for sum b, $v(b) = b^2/2$, so $v'(5) = 5 = -\nu^*$.

Why the Steps Work

Subtracting stationarity equations works because the same linear equality normal enters both coordinates. The completed square certifies a global minimum, not merely stationarity. The multiplier sign is checked against the value-function derivative using the stated $+\nu h$ convention.

A.6 Maximum-volume closed carton with fixed area

Problem

A closed rectangular carton has positive dimensions h, l, w, total surface area $10\,\mathrm{m}^2$, and volume $V = hlw$. Find the dimensions that maximize volume.

1. Write a minimization problem with the complete area formula.

$$\min -hlw, \qquad hl + hw + lw - 5 = 0, \qquad h, l, w > 0.$$

With $L = -hlw + \nu(hl + hw + lw - 5)$,

$$-lw + \nu(l + w) = 0, \quad -hw + \nu(h + w) = 0, \quad -hl + \nu(h + l) = 0.$$

2. Use symmetry without assuming it as a proof.

Multiply the three equations by h, l, w, respectively. Their objective terms all become $-hlw$. Subtracting the first two gives

$$\nu\{h(l + w) - l(h + w)\} = \nu w(h - l) = 0.$$

Because $w > 0$ and $\nu = lw/(l + w) > 0$, $h = l$. Repeating yields $h = l = w = a$.

3. Enforce area and evaluate volume.

$$3a^2 = 5, \qquad a = \sqrt{5/3} = 1.290994\,\mathrm{m},$$

$$V^* = \left(\frac{5}{3}\right)^{3/2} = 2.151657\,\mathrm{m}^3, \qquad \nu^* = \frac{a}{2} = 0.645497\,\mathrm{m}.$$

4. Prove that the stationary carton is globally best.

AM-GM applied to hl, hw, lw gives

$$\frac{hl + hw + lw}{3} \geq \big((hl)(hw)(lw)\big)^{1/3} = (hlw)^{2/3}.$$

Therefore $V \leq (5/3)^{3/2}$, with equality exactly when $hl = hw = lw$, hence $h = l = w$. If area is written as $2(hl + hw + lw) = S$, the multiplier for the original area equation is $\eta = a/4$, and $dV^*/dS = \eta$.

Why the Steps Work

Multiplying by dimensions creates identical volume terms that cancel cleanly. Positivity justifies division by νw; it must not be assumed silently. The AM-GM inequality supplies the missing global argument. Rescaling the area equation by two halves the corresponding multiplier, explaining why the sensitivity multiplier is $a/4$, not $a/2$.

A.7 Production capacities are inequalities, not compulsory usage

Problem

Maximize $P = 12x_1 + x_1^2 + 5x_2$, with raw-material capacity 100 and labor capacity 80:

$$3x_1 + 2x_2 \leq 100, \qquad 2x_1 + 4x_2 \leq 80, \qquad x_1, x_2 \geq 0.$$

Treat production quantities as continuous for this model.

1. Construct the feasible polygon.

The axis intercept limits are $x_1 \leq 100/3$ and $x_2 \leq 20$. The resource lines intersect at $(30, 5)$. The polygon vertices and profits are

(x_1, x_2)	Profit
$(0, 0)$	0
$(100/3, 0)$	1511.111111
$(30, 5)$	1285
$(0, 20)$	100

The objective is convex because $\nabla^2 P = \mathrm{diag}(2, 0) \succeq 0$. A maximum of a convex function over a compact polytope has at least one vertex maximizer: write any feasible point as a convex combination of vertices and apply the convexity inequality. Consequently,

$$x^* = \left(\frac{100}{3}, 0\right), \quad P^* = \frac{13600}{9} = 1511.111111.$$

2. Check the supposedly unused resource.

Raw material is exactly 100, but labor use is $200/3 = 66.666667 < 80$. The unused labor is $40/3$. This is allowed by a capacity upper bound.

3. Verify the multiplier signs.

Minimize $-P$, and assign multipliers $\mu_1, \mu_2, \mu_3, \mu_4$ to the two capacities and the two lower bounds. At the optimum, labor and the x_1 lower bound are inactive, so $\mu_2 = \mu_3 = 0$. Stationarity gives

$$-12 - 2x_1 + 3\mu_1 = 0 \;\Rightarrow\; \mu_1 = \frac{236}{9},$$

$$-5 + 2\mu_1 - \mu_4 = 0 \;\Rightarrow\; \mu_4 = \frac{427}{9} > 0.$$

4. Compare the different model that requires both capacities to be exhausted.

Replacing both inequalities by equalities forces $(30, 5)$, with profit 1285. For min $-P$, its equality multipliers are $(139/4, -129/8)$. The negative second multiplier is permitted for an equality, but would violate the sign requirement for the labor-capacity inequality.

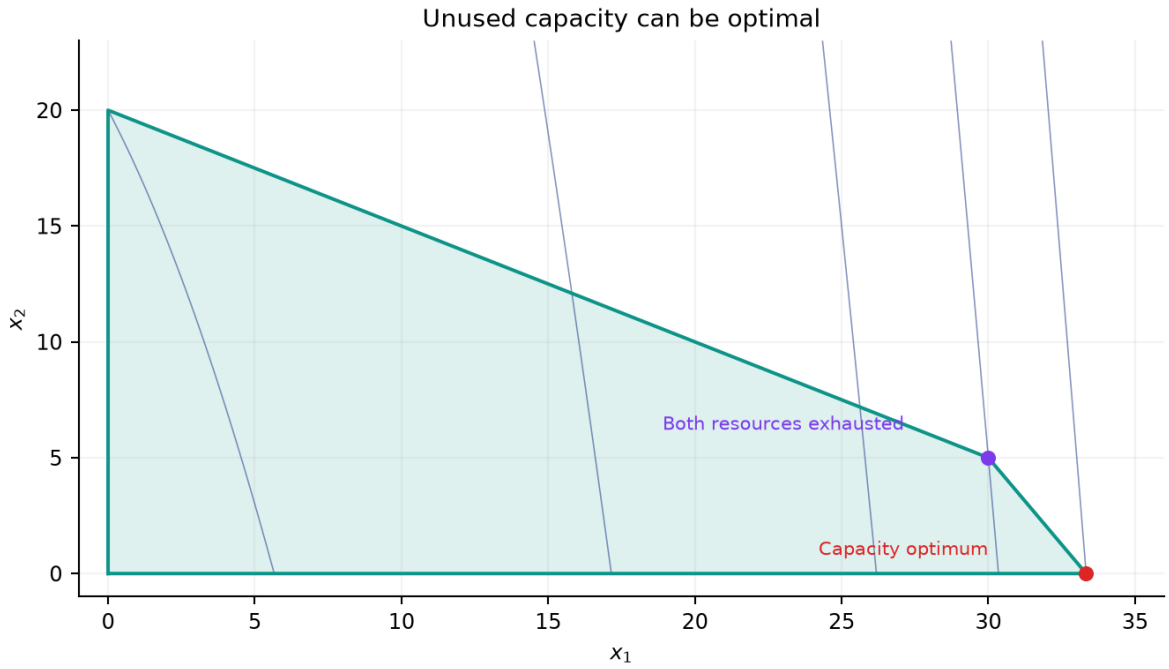


Figure A.2: The capacity model permits unused labor. Requiring both resource lines to hold at equality changes the optimization problem and lowers the attainable profit in this example.

Why the Steps Work

"Not exceeded" means an inequality. It does not mean every resource must be exhausted. The vertex comparison is a valid global argument here because the maximized objective is convex, even though this is not a convex optimization problem in the usual minimization sense. KKT signs diagnose why the equalities-only solution is not the optimum of the capacity model. If integer units are required, integrality must be imposed as a separate modelling condition.

A.8 Two nonlinear equalities and multiple feasible branches

Problem

Minimize $f = x_1^2 + x_2^2 + x_3^2$, subject to

$$h_1 = -x_1 + x_3 - 1 = 0, \qquad h_2 = x_1^2 - x_2^2 - x_1 = 0.$$

1. Form the correct Jacobian and stationarity equations.

$$J_h^T = \begin{bmatrix} -1 & 2x_1 - 1 \\ 0 & -2x_2 \\ 1 & 0 \end{bmatrix},$$

$$\begin{aligned} 2x_1 - \nu_1 + (2x_1 - 1)\nu_2 &= 0, \\ 2x_2 - 2\nu_2 x_2 &= 0, \\ 2x_3 + \nu_1 &= 0. \end{aligned}$$

There are two multipliers, one for each equality. The product $J_h^T \nu$ must use both columns with their own multipliers.

2. Reduce to a one-dimensional feasible domain.

Set $t = x_1$. Then $x_3 = t + 1$, $x_2^2 = t^2 - t \geq 0$, so $t \leq 0$ or $t \geq 1$, and

$$f(t) = t^2 + (t^2 - t) + (t + 1)^2 = 3t^2 + t + 1.$$

The unrestricted stationary point of this parabola is $t = -1/6$, which lies in $t \leq 0$. It produces

$$x^* = \left(-\frac{1}{6}, \pm\frac{\sqrt{7}}{6}, \frac{5}{6}\right), \quad f^* = \frac{11}{12}, \quad (\nu_1, \nu_2) = \left(-\frac{5}{3}, 1\right).$$

3. Classify the branch endpoints instead of ignoring them.

On $t \geq 1$, $f'(t) = 6t + 1 > 0$, so the branch minimum occurs at $t = 1$, giving $(1, 0, 2)$, value 5. At $t = 0$, the point $(0, 0, 1)$ has value 1; nearby feasible points on the negative branch have smaller value. Its tangent curvature is negative, so it is a local maximum on that branch.

Point	Objective	Classification
$(-1/6, \pm\sqrt{7}/6, 5/6)$	$11/12$	Global minima
$(0, 0, 1)$	1	Branch-local maximum
$(1, 0, 2)$	5	Nonglobal local minimum

Why the Steps Work

The constraint $x_2^2 = t(t-1)$ splits the feasible set into separated branches. Solving the reduced quadratic over all real t would miss that domain restriction, and checking only one sign of x_2 would discard a second global minimizer. The final classification uses the restricted domains, not just the unconstrained shape of $3t^2 + t + 1$.

A.9 A linear program and zero multipliers at an active boundary

Problem

Maximize $4x_1 + 3x_2$ subject to

$$g_1 = 2x_1 + x_2 - 10 \le 0, \quad g_2 = x_1 + 2x_2 - 8 \le 0, \quad g_3 = -x_1 \le 0, \quad g_4 = -x_2 \le 0.$$

Minimize the negative objective. If both resource constraints are active, solve

$$2x_1 + x_2 = 10, \qquad x_1 + 2x_2 = 8,$$

to obtain $x = (4, 2)$. Since both coordinates are positive, $\mu_3 = \mu_4 = 0$. Stationarity becomes

$$-4 + 2\mu_1 + \mu_2 = 0, \qquad -3 + \mu_1 + 2\mu_2 = 0.$$

Thus $\mu_1 = 5/3$, $\mu_2 = 2/3$, both positive. Feasibility, dual feasibility, complementary slackness, and stationarity all hold. The minimized objective is affine and the feasible set is convex, so this is a global certificate.

Vertex	$(0,0)$	$(5,0)$	$(4,2)$	$(0,4)$
Profit	0	20	22	12

For comparison, minimize $x_1^2 + x_2^2$ with $2x_1 + x_2 \le 10$, $x_1 + x_2 \le 4$, and $x_i \ge 0$. At $(0,0)$, the two lower bounds are active but the gradient is zero. All multipliers may be zero, and $f \ge 0$ gives a direct global certificate.

Why the Steps Work

Negating a maximized objective is necessary before applying the stated minimization KKT convention. Strict positivity of the coordinates makes their lower bounds inactive, forcing those multipliers to zero. Conversely, a constraint being active only means its slack is zero; it does not require a strictly positive multiplier. The vertex table independently confirms the linear-program optimum.

A.10 Quadratic maximization: why every boundary matters

Problem

Maximize $Q = 4x_1^2 + 3x_2^2$ over the same polygon: $2x_1 + x_2 \le 10$, $x_1 + 2x_2 \le 8$, $x_1, x_2 \ge 0$.

This maximizes a convex quadratic, so minimizing $-Q$ is nonconvex. KKT conditions are necessary at regular local optima, but are not a global sufficiency theorem here.

1. Check the global vertex comparison.

Vertex	$(0,0)$	$(5,0)$	$(4,2)$	$(0,4)$
Q	0	100	76	48

Convexity of Q ensures that some vertex attains the maximum over this compact polygon. Therefore $x^* = (5,0)$, $Q^* = 100$.

2. Inspect the tempting two-resource intersection.

At $(4,2)$, lower bounds are inactive. Stationarity of $-Q$ requires

$$2\mu_1 + \mu_2 = 32, \qquad \mu_1 + 2\mu_2 = 12,$$

so $(\mu_1, \mu_2) = (52/3, -8/3)$. The negative second multiplier violates dual feasibility. A feasible point with these multipliers is not a KKT point of the original inequality problem.

3. Classify the one-resource stationary candidates.

If only $g_1 = 0$ is active, the candidate is $(15/4, 5/2)$, but $x_1 + 2x_2 = 35/4 > 8$, so it is infeasible. If only $g_2 = 0$ is active, the candidate is

$$x = \left(\frac{24}{19}, \frac{64}{19}\right), \qquad \mu_2 = \frac{192}{19} > 0.$$

It is feasible and satisfies first-order KKT. However, the tangent direction $d = (2, -1)$ has $\nabla g_2^T d = 0$ and

$$d^T \nabla^2(-Q) d = (2,-1) \begin{bmatrix} -8 & 0 \\ 0 & -6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = -38 < 0.$$

Both small tangent signs are feasible there. Negative tangent curvature rules out a minimum of $-Q$, hence a maximum of Q.

4. Include the active lower bound at the true optimum.

At $(5,0)$, $g_1 = 0$, $g_4 = 0$, while $g_2 < 0$, $g_3 < 0$. Thus $\mu_2 = \mu_3 = 0$, and

$$(-40, 0) + \mu_1(2,1) + \mu_4(0,-1) = 0 \quad \Rightarrow \quad \mu_1 = 20,\ \mu_4 = 20.$$

All KKT signs are valid. At $(0,4)$, there is another valid local KKT candidate with $\mu_2 = \mu_3 = 12$, but its value 48 is smaller than 100. The origin has zero gradient and zero multipliers, yet moving into the feasible region increases Q; it is not a maximizer.

Why the Steps Work

Ignoring nonnegativity is not harmless because the actual maximum uses $x_2 = 0$. A negative multiplier rejects the two-resource intersection. A positive-multiplier stationary point can still fail second-order conditions in a nonconvex minimization. The final global conclusion comes from a justified vertex comparison, not merely from solving stationarity equations.

A.11 A circle-constrained problem and its exact dual certificate

Problem

Minimize $f = (x_1 - 2)^2 + (x_2 + 3)^2 + 1$, subject to $x_1^2 + x_2^2 = 8$ and $x_1 + x_2 \leq 4$, with $x \in \mathbb{R}^2$.

1. Identify the geometry and activity.

Let $c = (2, -3)$, so $f = \|x - c\|^2 + 1$. On the circle, $\|x\| = \sqrt{8}$, and Cauchy–Schwarz gives

$$x_1 + x_2 \leq \sqrt{2}\,\|x\| = 4.$$

The inequality is redundant on the circle, although it is active at $(2, 2)$. The closest circle point to c lies along c:

$$x^* = \frac{\sqrt{8}}{\sqrt{13}}(2, -3) = (1.568929, -2.353394).$$

Its sum is negative, so the inequality is strictly inactive and $\mu^* = 0$.

2. Evaluate and verify stationarity.

The Lagrangian is

$$L = (x_1 - 2)^2 + (x_2 + 3)^2 + 1 + \nu(x_1^2 + x_2^2 - 8) + \mu(x_1 + x_2 - 4).$$

With $\mu = 0$, stationarity gives $(1 + \nu)x = c$. Therefore

$$\nu^* = \sqrt{13/8} - 1 = 0.2747548784, \qquad f^* = (\sqrt{13} - \sqrt{8})^2 + 1 = 22 - 4\sqrt{26} = 1.6039219456.$$

3. Derive the dual function by completing the square.

Expand the Lagrangian:

$$L = (1 + \nu)(x_1^2 + x_2^2) + (\mu - 4)x_1 + (\mu + 6)x_2 + 14 - 8\nu - 4\mu.$$

For $\nu > -1$, minimizing each scalar quadratic yields

$$q(\nu, \mu) = 14 - 8\nu - 4\mu - \frac{(\mu - 4)^2 + (\mu + 6)^2}{4(1 + \nu)}, \qquad \mu \geq 0.$$

For $\nu < -1$, negative curvature makes the infimum $-\infty$. At $\nu = -1$, both linear coefficients would have to vanish for finiteness, requiring $\mu = 4$ and $\mu = -6$ simultaneously. This is impossible, so the infimum is again $-\infty$.

4. Match the lower bound to the feasible objective.

At $\mu = 0$, $q = 14 - 8\nu - 13/(1 + \nu)$. Setting

$$\frac{\partial q}{\partial \nu} = -8 + \frac{13}{(1 + \nu)^2} = 0$$

gives $\nu^* = \sqrt{13/8} - 1$ in the finite domain. Substitution produces $q(\nu^*, 0) = f(x^*)$. Weak duality now proves that no feasible point can be better.

Why the Steps Work

The geometric projection solves the primal problem directly. Checking the inequality identifies its multiplier as zero rather than assuming activity. Completing the square determines the dual's finite domain. Matching a feasible objective value and a dual lower bound gives a global certificate even though an equality circle is a nonconvex feasible set; no blanket strong-duality assumption is needed.

A.12 Projection onto a parabola region: complete active-set calculation

Problem

Minimize $(x_1 - 3)^2 + (x_2 - 4)^2$, subject to

$$g_1 = x_1 + x_2 - 10 \le 0, \quad g_2 = x_1^2 - x_2 \le 0, \quad g_3 = -x_1 \le 0, \quad g_4 = -x_2 \le 0.$$

The objective is strictly convex; g_1, g_3, g_4 are affine and g_2 is convex. A feasible KKT point therefore certifies the unique global minimizer.

1. Form stationarity before choosing active constraints.

$$\begin{aligned} 2(x_1 - 3) + \mu_1 + 2\mu_2 x_1 - \mu_3 &= 0, \\ 2(x_2 - 4) + \mu_1 - \mu_2 - \mu_4 &= 0. \end{aligned}$$

First search for a solution with positive coordinates, so $\mu_3 = \mu_4 = 0$. This is a hypothesis that must be checked in the final answer.

2. Reject the unconstrained candidate.

If $\mu_1 = \mu_2 = 0$, then $x = (3, 4)$, but $g_2 = 9 - 4 = 5 > 0$. The desired unconstrained target lies below the parabola.

3. Solve the parabola-active, line-inactive case.

Set $\mu_1 = 0$, $x_2 = x_1^2$. The second stationarity equation gives $\mu_2 = 2(x_1^2 - 4)$. Substituting into the first and dividing by two gives

$$x_1 - 3 + 2x_1(x_1^2 - 4) = 0 \quad \Rightarrow \quad 2x_1^3 - 7x_1 - 3 = 0.$$

All three real roots must be checked:

x_1	$x_2 = x_1^2$	$\mu_2 = 2(x_2 - 4)$	Decision
-1.600956	2.56306	-2.87388	Reject
-0.455589	0.207562	-7.584877	Reject
2.056545	4.229379	0.458757	Accept

The accepted root gives

$$x^* = (2.0565452922,\ 4.2293785387), \quad \mu_2^* = 0.4587570774, \quad f^* = 0.9427212998.$$

Here $x_1 + x_2 = 6.285923831 < 10$, so setting $\mu_1 = 0$ was consistent. Both coordinates are positive, so the omitted lower-bound multipliers are indeed zero.

4. Examine the other two resource-activity cases.

With only the line active, subtract the stationarity equations to get $x_1 - x_2 = -1$. Together with sum 10 this gives $(4.5, 5.5)$, $\mu_1 = -3$. It violates dual feasibility and the parabola constraint.

With both active, $x_2 = 10 - x_1 = x_1^2$, hence

$$x_1 = \frac{-1 \pm \sqrt{41}}{2}, \qquad x_2 = 10 - x_1.$$

At the positive root, $x_1 \approx 2.701562$, $x_2 \approx 7.298438$. Stationarity gives

$$\mu_2 = \frac{2x_2 - 2x_1 - 2}{2x_1 + 1} \approx 1.123475, \qquad \mu_1 = 8 - 2x_2 + \mu_2 \approx -5.473401 < 0.$$

Reject it. The negative root already violates $x_1 \geq 0$. No further lower-bound case search is required to prove the result, because a valid KKT point of this strictly convex problem has already been found and verified against *all* original constraints.

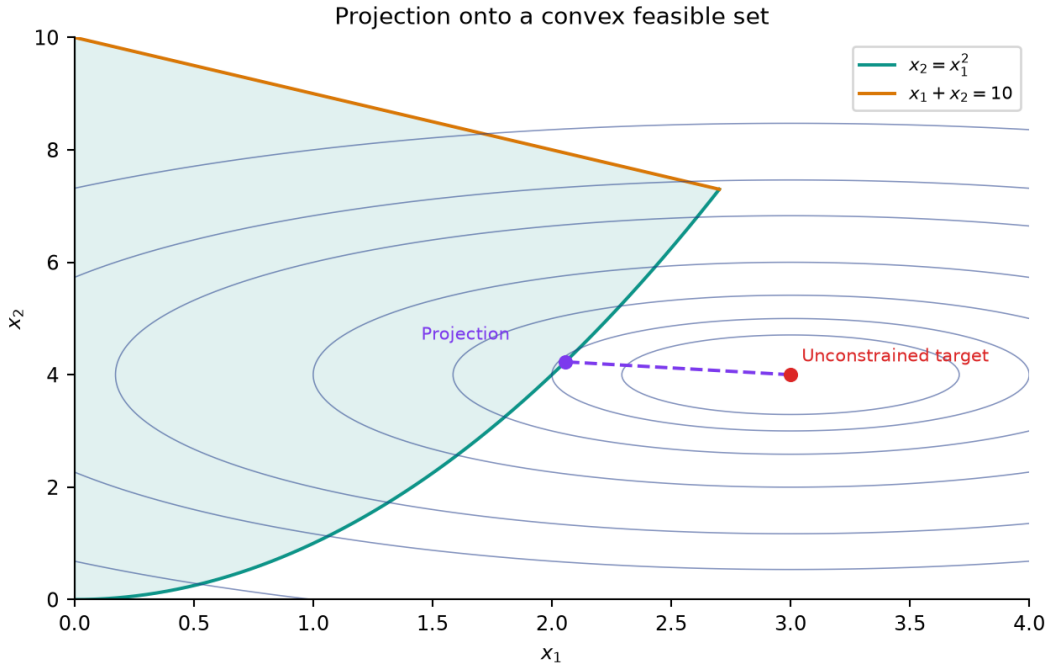


Figure A.3: The constrained solution is the closest feasible point to $(3, 4)$. Only the parabola boundary is active at the solution.

Why the Steps Work

Complementary slackness converts activity hypotheses into equations. The cubic comes from substituting both the active boundary and the multiplier relation, not from numerical guessing. Rejecting roots uses multiplier signs and the original variable bounds. Strict convexity plus a verified KKT point supplies global uniqueness, so it is unnecessary to enumerate every remaining boundary after obtaining this certificate.

A.13 Quadratic penalties and their finite-parameter bias

Problem

Solve $\min x$ subject to $x \geq 5$, using the penalty $F_\rho(x) = x + \rho \max(5 - x, 0)^2$, and explain its limit.

For $x \geq 5$, the penalty is zero and $F_\rho = x$, whose feasible minimum is 5. For $x < 5$,

$$F'_\rho = 1 - 2\rho(5 - x), \qquad F''_\rho = 2\rho > 0.$$

The unique exterior stationary point is

$$x_\rho = 5 - \frac{1}{2\rho} < 5, \qquad F_\rho(x_\rho) = 5 - \frac{1}{4\rho} < 5.$$

Thus it beats every feasible point in the penalized objective and is the global penalized minimizer. The violation is $5 - x_\rho = 1/(2\rho)$. To make it at most 10^{-3}, choose $\rho \geq 500$, provided the subproblem itself is solved accurately.

A radial extension.

For the explicitly defined example $\min \|x\|^2$ subject to $\|x\|^2 \geq 2$, let $s = \|x\|^2 \geq 0$. The quadratic penalty is

$$F_\rho(s) = s + \rho \max(2 - s, 0)^2.$$

If $\rho \geq 1/4$, its minimum occurs at $s_\rho = 2 - 1/(2\rho)$, which is infeasible for every finite ρ. If $0 < \rho < 1/4$, the stationary value would be negative, so the domain $s \geq 0$ forces $s_\rho = 0$. All directions with the minimizing radius are equivalent.

Why the Steps Work

The squared positive-part penalty must be differentiated piecewise. Comparing the exterior stationary value with the feasible boundary proves which branch actually minimizes the penalized objective. The factor $1/(2\rho)$ comes from the stated coefficient ρ, not $\rho/2$; changing that convention changes the numerical formula. A small penalty bias is not exact feasibility.

A.14 Augmented Lagrangian: learning the constraint force

Problem

Minimize x^2 subject to $h = x - 1 = 0$. Use $\nu_0 = 0$, $\rho_0 = 1$, and double ρ after each multiplier update.

The augmented objective is

$$\mathcal{L}_\rho = x^2 + \nu(x-1) + \frac{\rho}{2}(x-1)^2.$$

Set its derivative to zero:

$$2x + \nu + \rho(x-1) = 0 \quad \Rightarrow \quad x_{k+1} = \frac{\rho_k - \nu_k}{\rho_k + 2}.$$

The multiplier update is $\nu_{k+1} = \nu_k + \rho_k(x_{k+1} - 1)$.

At $k = 0$: $x_1 = 1/3$, $h(x_1) = -2/3$, and $\nu_1 = 0 + 1(-2/3) = -2/3$. Double ρ to 2. At $k = 1$:

$$x_2 = \frac{2 - (-2/3)}{4} = \frac{2}{3}, \qquad \nu_2 = -\frac{2}{3} + 2\left(-\frac{1}{3}\right) = -\frac{4}{3}.$$

k	ρ_k	ν_k	x_{k+1}	$h(x_{k+1})$	ν_{k+1}
0	1	0	0.333333	-0.666667	-0.666667
1	2	-0.666667	0.666667	-0.333333	-1.333333
2	4	-1.333333	0.888889	-0.111111	-1.777778
3	8	-1.777778	0.977778	-0.022222	-1.955556
4	16	-1.955556	0.997531	-0.002469	-1.995062
5	32	-1.995062	0.999855	-0.000145	-1.99971
6	64	-1.99971	0.999996	-4.401×10^{-6}	-1.999991

At the exact solution, stationarity is $2(1) + \nu^* = 0$, so $\nu^* = -2$. If this multiplier is used, the augmented subproblem returns $x = (\rho + 2)/(\rho + 2) = 1$ for any positive ρ.

Even holding ρ fixed can work for this example. Substituting the minimizer into the update gives

$$\nu_{k+1} + 2 = \frac{2}{\rho + 2}(\nu_k + 2), \qquad x_{k+1} - 1 = -\frac{\nu_k + 2}{\rho + 2}.$$

For $\rho = 1$, the multiplier error shrinks by exactly $2/3$ per outer iteration, while the pure penalty solution remains at $1/3$.

Why the Steps Work

Differentiate with ν and ρ fixed inside the subproblem, then update the multiplier using the new constraint residual. The linear term learns the nonzero stationarity force required at the constrained solution. The explicit recurrence proves bounded-penalty convergence for this example; it does not claim exact feasibility after a finite number of arbitrary multiplier updates.

A.15 Scalar barriers and two central-path calculations

Part I: approach to a boundary

Minimize $f(x) = x$ subject to $x \geq 1$. Compare a quadratic penalty with a log barrier.

With the convention $x + (\rho/2)(1-x)_+^2$, the exterior derivative is $1 - \rho(1-x)$, giving $x_\rho = 1 - 1/\rho$. With the log barrier,

$$F_t = x - \frac{1}{t}\log(x-1), \quad x > 1, \qquad F_t' = 1 - \frac{1}{t(x-1)} = 0,$$

so $x_t = 1 + 1/t$, which is strictly feasible. The multiplier estimate is $\mu_t = 1/[t(x_t - 1)] = 1$, exactly the optimal multiplier. The gap is $f(x_t) - f^* = 1/t$, equal to the one-constraint bound m/t.

$\rho = t$	Penalty $1 - 1/\rho$	Barrier $1 + 1/t$
1	0	2
2	0.5	1.5
5	0.8	1.2
10	0.9	1.1
100	0.99	1.01
1000	0.999	1.001

Part II: a linear objective on a five-constraint polygon

Minimize $x_1 + x_2$ with $0 \leq x_i \leq 2$ and $x_1 + x_2 \leq 3$.

The constraints in $a_i^T x \leq b_i$ form are

Constraint	a_i^T	b_i
$x_1 \leq 2$	$(1, 0)$	2
$x_2 \leq 2$	$(0, 1)$	2
$x_1 \geq 0$	$(-1, 0)$	0
$x_2 \geq 0$	$(0, -1)$	0
$x_1 + x_2 \leq 3$	$(1, 1)$	3

The unique objective minimizer is $(0, 0)$, with value zero. The barrier problem is

$$F_t = t(x_1 + x_2) - \log(2 - x_1) - \log(2 - x_2) - \log x_1 - \log x_2 - \log(3 - x_1 - x_2).$$

Its Hessian is positive definite on the strict interior. Symmetry and uniqueness imply $x_1 = x_2 = s$. One partial derivative yields

$$t + \frac{1}{2-s} - \frac{1}{s} + \frac{1}{3-2s} = 0, \qquad 0 < s < 1.5.$$

The derivative of the left side with respect to s is positive, and its endpoint limits have opposite signs, so a bracketed root solve has exactly one root.

t	$s(t)$	Objective gap	Bound $5/t$
0.1	0.680936	1.361872	50
1	0.467911	0.935822	5
10	0.091919	0.183837	0.5
100	0.009917	0.019834	0.05
1000	0.000999	0.001998	0.005

The original objective gap is $2s$, while the certified ideal barrier bound is $5/t$. This bound need not be tight. The slack products $\mu_j s_j$ all equal $1/t$.

Part III: change the objective, preserve the constraints

Minimize $(x_1 - 5)^2 + (x_2 - 5)^2$ over the same polygon.

The solution is $(1.5, 1.5)$, with value 24.5. Its sum constraint is active; the other inequalities are slack. Stationarity gives $(-7, -7) + \mu_5(1, 1) = 0$, so $\mu_5 = 7$.

For the barrier path, symmetry again gives $x_1 = x_2 = s$, but the stationarity equation becomes

$$2t(s - 5) + \frac{1}{2 - s} - \frac{1}{s} + \frac{1}{3 - 2s} = 0.$$

t	$s(t)$	Objective gap	Bound $5/t$
0.1	0.9503	8.300144	50
1	1.418665	1.151927	5
10	1.492737	0.101785	0.5
100	1.499285	0.010018	0.05
1000	1.499929	0.001	0.005

Here $s \to 1.5$ from below. In the linear problem, $s \to 0$. Both paths are straight diagonal segments by symmetry, even though central paths in general can be curved.

Why the Steps Work

The barrier is defined only on positive slacks. Symmetry is used only after uniqueness has been justified, so it does not discard asymmetric alternatives. The logarithmic chain rule determines every derivative sign. The gap bound counts all five inequalities, including those inactive at the final optimum, because each contributes $1/t$ to perturbed complementarity. A different objective changes the limiting active set even on the same feasible polygon.

A.16 Unconstrained Newton: exact model equations and numerical accuracy

Part I: an exponential objective

Minimize $f(x) = e^x - 2x$, starting at $x_0 = 0$.

The derivatives are $f' = e^x - 2$, $f'' = e^x > 0$. Thus the unique stationary point is the global minimum,

$$x^* = \log 2, \qquad f^* = 2 - 2\log 2 = 0.6137056389.$$

The Newton update is

$$d_k = -\frac{e^{x_k} - 2}{e^{x_k}} = -1 + 2e^{-x_k}, \qquad x_{k+1} = x_k - 1 + 2e^{-x_k}.$$

At $x_0 = 0$, $g = -1$, $H = 1$, so $d_0 = 1$. The model is $q(d) = 1 - d + d^2/2$, minimized at $d = 1$. This predicts decrease $1/2$, whereas the actual first-step decrease is $1 - (e - 2) \approx 0.281718$. Model decrease and actual decrease are different away from the solution.

$$\frac{\delta_k^2}{2} = \frac{(e^{x_k} - 2)^2}{2e^{x_k}}.$$

k	x_k	Position error	$\delta^2/2$	Objective gap
0	0	0.693147	0.5	0.386294
1	1	0.306853	0.0949	0.104576
2	0.735759	0.042612	0.001816	0.001842
3	0.694042	0.000895	8.012×10^{-7}	8.015×10^{-7}
4	0.693148	4.005×10^{-7}	1.604×10^{-13}	1.605×10^{-13}
5	0.693147	8.016×10^{-14}	0	0

Near the solution, the model decrease and true objective gap agree closely. The position error shrinks approximately quadratically once the iterates are close.

Part II: curvature rescaling in two variables

Minimize $x_1^2 + 4x_2^2$ from $x_0 = (2, 1)$.

Here $g_0 = (4, 8)^T$ and $H = \text{diag}(2, 8)$. Solve

$$\begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -8 \end{bmatrix} \quad \Rightarrow \quad d = (-2, -1), \qquad x_1 = (0, 0).$$

For steepest descent, $d = -g$. Exact line search on this quadratic gives

$$\alpha_0 = \frac{g_0^T g_0}{g_0^T H g_0} = \frac{80}{544} = \frac{5}{34}, \quad x_1 = \left(\frac{24}{17}, -\frac{3}{17}\right), \quad f(x_1) = \frac{36}{17}.$$

Recomputing each exact line search requires 12 gradient-descent iterations to reach $f < 10^{-6}$, while Newton takes one full step.

Why the Steps Work

Newton applies root finding to f', not to f. The model Hessian rescales steep directions more strongly, which explains the one-step quadratic result. The decrement is computed before taking the step and describes the current model. Its agreement with the true gap is local for the exponential objective, not an identity valid at every iterate.

A.17 Backtracking rescues an otherwise divergent Newton iteration

Problem

Minimize $f(x) = \sqrt{1+x^2}$ from $x_0 = 1.5$. Use Armijo parameters $c = 1/4$, $\beta = 1/2$.

1. Derive the full-step iteration and its failure.

$$f' = \frac{x}{\sqrt{1+x^2}}, \qquad f'' = (1+x^2)^{-3/2}, \qquad d = -x(1+x^2).$$

A full update gives $x_+ = x + d = -x^3$. Therefore $|x_+| = |x|^3$: starts with $|x_0| < 1$ converge to zero, starts with $|x_0| > 1$ diverge, and $x_0 = \pm 1$ forms a two-cycle.

k	$x_0 = 0.5$	$x_0 = 1.5$
0	0.5	1.5
1	-0.125	-3.375
2	0.001953	38.443359
3	-7.451×10^{-9}	-56815.128662

2. Compute the first search direction and allowed decrease.

At $x = 1.5$,

$$f = \sqrt{3.25} = 1.802775638, \qquad d = -4.875, \quad \delta^2 = -f'd = 4.056245185.$$

For each trial α, compare the actual trial value with $f(x) - \alpha\delta^2/4$:

Step α	Trial point	Trial f	Armijo limit	Accept?
1	-3.375	3.520032	0.788714	No
0.5	-0.9375	1.370732	1.295745	No
0.25	0.28125	1.038798	1.54926	Yes

The first acceptable trial is $\alpha = 1/4$, giving $x_1 = 0.28125$. The half step is rejected even though it reduces f: its reduction is insufficient relative to the directional prediction.

3. Continue from the accepted iterate.

k	x_k	Accepted α	$\delta^2/2$	Objective gap
0	1.5	0.25	2.028123	0.802776
1	0.28125	1	0.041085	0.038798
2	-0.022247	1	0.000248	0.000247
3	1.101×10^{-5}	1	6.062×10^{-11}	6.062×10^{-11}
4	0	1	0	0

After the damped first step, full steps are accepted, and the iterates approach zero rapidly. The minimum is $f(0) = 1$.

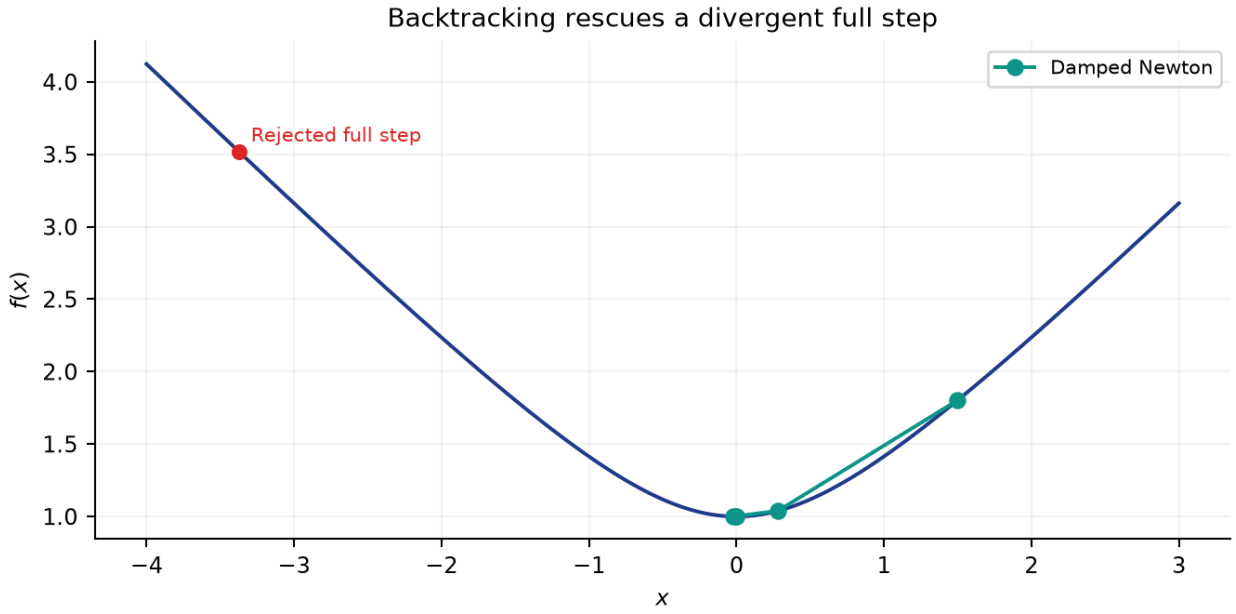


Figure A.4: The rejected full step overshoots to a high objective value. Backtracking moves into a region where the local Newton model is reliable.

Why the Steps Work

Positive curvature makes the direction locally descending, but does not make the full step safe. Armijo compares actual decrease with a fraction of the model's first-order prediction, so it can reject a decreasing but insufficient half step. The accepted short step restores a useful local regime; subsequent full steps exploit fast local convergence.

A.18 Feasible equality Newton on an exact quadratic

Problem

Minimize $x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$, starting at $x_0 = (1, 0)$.

The start is feasible. With $A = [1\ 1]$, $b = 1$, $g_0 = (2, 0)$, $H = 2I$, solve

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \nu_+ \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}.$$

The equations are $2d_1 + \nu_+ = -2$, $2d_2 + \nu_+ = 0$, and $d_1 + d_2 = 0$. Substitute $d_2 = -d_1$ into the second equation, obtaining $\nu_+ = 2d_1$. Then $4d_1 = -2$, so

$$d = (-1/2, 1/2), \qquad \nu_+ = -1, \qquad x_1 = (1/2, 1/2).$$

Feasibility is $1/2 + 1/2 = 1$. Stationarity is $(1, 1) + (-1)(1, 1) = 0$. The decrement is

$$\delta^2 = d^T(2I)d = 1, \qquad \delta^2/2 = 1/2 = f(x_0) - f(x_1).$$

All equality-feasible steps remain on the same line. Strict convexity certifies the unique global solution, $f^* = 1/2$.

Why the Steps Work

The zero in the last row's right-hand side means the direction must preserve current feasibility. Eliminating d_2 uses that tangent equation before solving stationarity. Since the objective model is exact, the predicted decrease equals the true gap and a full step solves the problem in one iteration. The returned -1 is a new model multiplier, not an increment to add blindly.

A.19 Feasible equality Newton for an exponential objective

Problem

Minimize $f = e^{x_1} + e^{x_2}$ subject to $x_1 + x_2 = 2$, starting at $(2, 0)$.

1. Derive and solve the first KKT system.

Let $a = e^2 = 7.389056099$. Then

$$\begin{bmatrix} a & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \nu_+ \end{bmatrix} = \begin{bmatrix} -a \\ -1 \\ 0 \end{bmatrix}.$$

The last equation gives $d_2 = -d_1$. The second gives $\nu_+ = -1 + d_1$. Substituting into the first gives

$$(a+1)d_1 = 1 - a, \qquad d_1 = -\frac{a-1}{a+1} = -0.761594156, \qquad d_2 = 0.761594156.$$

Thus $\nu_+ = -1.761594156$, and the full step reaches $(1.238405844, 0.761594156)$.

2. Recompute derivatives at each new point.

$$\nabla f = (e^{x_1}, e^{x_2})^T, \qquad H = \operatorname{diag}(e^{x_1}, e^{x_2}).$$

Because the Hessian changes with the iterate, the first quadratic model is not exact globally. Repeated KKT solves with Armijo $c = 1/4$ accept full steps for this start:

k	x_1	x_2	$f(x_k)$	$\delta^2/2$	Objective gap
0	2	0	8.389056	2.432934	2.952492
1	1.238406	0.761594	5.591797	0.153078	0.155233
2	1.004416	0.995584	5.436617	5.302×10^{-5}	5.302×10^{-5}
3	1	1	5.436564	0	0
4	1	1	5.436564	0	0

At the solution, $x^* = (1, 1)$, $f^* = 2e$, and $\nu^* = -e$. To verify the global result, write $x = (1 + u, 1 - u)$:

$$f = 2e \cosh u \geq 2e,$$

with equality only at $u = 0$. In the same reduced coordinate, Newton is $u_+ = u - \tanh u$, explaining the rapidly shrinking imbalance.

Why the Steps Work

The constraint equation reduces a three-unknown KKT system to one scalar direction. Both coordinates always sum to two because $d_1 + d_2 = 0$. The global certificate follows from $\cosh u \geq 1$, independently of the numerical convergence trace. A tiny objective error can occur before all coordinates reach machine precision; the table should be interpreted with the displayed rounding and stopping rule in mind.

A.20 Economic dispatch, marginal prices, and an active capacity limit

Problem

Two generators supply 300 MW. Minimize the hourly cost

$$C(P) = 0.01P_1^2 + 2P_1 + 0.02P_2^2 + 1.5P_2, \qquad P_1 + P_2 = 300.$$

Start at equal sharing $(150, 150)$, then analyze a 10 MW demand increase and a 180 MW limit on generator 1.

1. Form the derivatives in physical units.

$$g = \begin{bmatrix} 0.02P_1 + 2 \\ 0.04P_2 + 1.5 \end{bmatrix}, \qquad H = \begin{bmatrix} 0.02 & 0 \\ 0 & 0.04 \end{bmatrix}.$$

At the start, $g = (5, 7.5)^T$ dollars per MWh, while $C = 1200$ dollars per hour. Generator 2 has the higher marginal cost, suggesting a shift toward generator 1.

2. Solve the demand-preserving Newton system.

$$\begin{bmatrix} 0.02 & 0 & 1 \\ 0 & 0.04 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \nu_+ \end{bmatrix} = \begin{bmatrix} -5 \\ -7.5 \\ 0 \end{bmatrix}.$$

Since $d_2 = -d_1$, subtracting the first two equations gives

$$0.02d_1 - 0.04d_2 = 2.5 \quad \Rightarrow \quad 0.06d_1 = 2.5.$$

Therefore $d = (125/3, -125/3)$, and

$$P^* = \left(\frac{575}{3}, \frac{325}{3}\right) = (191.666667, 108.333333) \text{ MW}.$$

The common marginal cost is $35/6 = 5.833333$ dollars per MWh, so $\nu^* = -35/6$. The cost is 1147.916667 dollars per hour, a saving of 52.083333. The positive-definite quadratic and exact demand equation make this one full Newton step globally optimal.

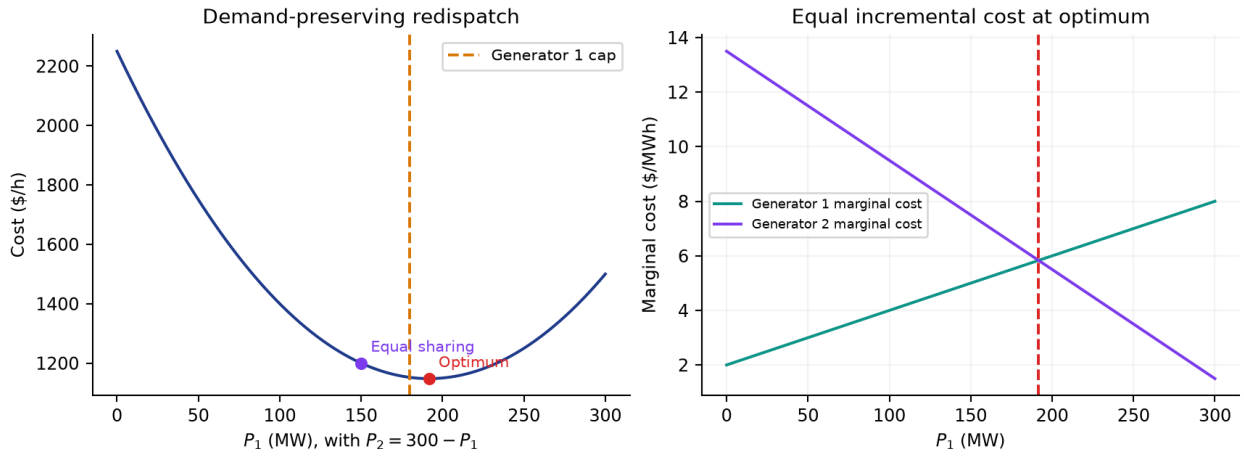


Figure A.5: Redispatch preserves demand and moves toward equal marginal costs. The minimum occurs where both generators have the same incremental cost.

3. Use sensitivity and quantify its approximation error.

For demand D, eliminate $P_2 = D - P_1$ or equalize marginal costs:

$$0.02P_1 + 2 = 0.04(D - P_1) + 1.5 \quad \Rightarrow \quad P_1(D) = \frac{2D}{3} - \frac{25}{3}.$$

The optimal price is $\pi(D) = D/75 + 11/6$. Thus

$$C^*(310) - C^*(300) = \int_{300}^{310} \pi(D)\, dD = 10\left(\frac{35}{6}\right) + \frac{10^2}{150} = 59.$$

The first-order multiplier estimate is 58.333333, low by $2/3$. Demand raises marginal costs along the reoptimized allocation, accounting for the positive second-order correction.

4. Impose generator 1's limit.

The unconstrained dispatch violates $P_1 \leq 180$. The capped solution is $(180, 120)$, with cost 1152. For

$$L = C + \nu(P_1 + P_2 - 300) + \mu(P_1 - 180),$$

stationarity gives $5.6 + \nu + \mu = 0$, $6.3 + \nu = 0$, hence

$$\nu = -6.3, \qquad \mu = 0.7 \geq 0.$$

Increasing the cap by a small amount reduces optimal cost at approximately 0.7 dollars per hour per extra MW of capacity, while this active set remains unchanged.

Why the Steps Work

The last KKT row encodes conservation of power, so redispatch must increase one unit exactly as much as it decreases the other. Equal marginal cost is a consequence of stationarity, not an independent guess. The demand multiplier is a first-order sensitivity and misses the quadratic cost increase for a finite demand change. A capacity bound changes the active set and introduces a nonnegative multiplier, so equality-only Newton cannot simply accept the previous dispatch.

A.21 Infeasible-start Newton can increase the objective correctly

Problem

Minimize $x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$, starting at $x_0 = (0, 0)$, $\nu_0 = 0$.

The residuals are $r_d = (0, 0)^T$, $r_p = -1$. The primal-dual increment system is

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \nu \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Its first two equations give $\Delta x_1 = \Delta x_2 = -\Delta\nu/2$. The last requires their sum to be one. Consequently,

$$\Delta x = (1/2, 1/2), \qquad \Delta\nu = -1.$$

A full step reaches $x_1 = (1/2, 1/2)$, $\nu_1 = -1$, with zero primal and dual residuals. The combined residual norm falls from 1 to zero, although the objective rises from zero to $1/2$.

For a half step, $x = (1/4, 1/4)$, $\nu = -1/2$. Then $r_d = (1/2, 1/2) - (1/2, 1/2) = 0$, while $r_p = -1/2$, exactly $(1 - \alpha)r_{p,0}$. Its residual norm is $1/2$.

Why the Steps Work

Zero gradient at the initial point does not solve a constrained problem when that point is infeasible. The nonzero last right-hand-side entry asks the step to cancel the constraint violation. Because both objective gradient and constraints are affine here, the KKT residual linearization is exact and a full Newton step solves the equations. Objective comparison against an infeasible point would reject precisely the step needed to reach a usable design.

A.22 Infeasible-start Newton with a nonlinear dual residual

Problem

Minimize $e^{x_1} + e^{x_2}$ subject to $x_1 + x_2 = 2$, starting at $(0, 1)$, $\nu_0 = 0$. Use residual backtracking with $c = 1/4$, $\beta = 1/2$.

1. Construct residuals and their Jacobian.

$$r_p = -1, \quad r_d = (1, e)^T, \quad K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & e & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The initial residual norm is $\sqrt{1 + e^2 + 1} = 3.064156670$. Solve

$$K \begin{bmatrix} d_1 \\ d_2 \\ \Delta\nu \end{bmatrix} = \begin{bmatrix} -1 \\ -e \\ 1 \end{bmatrix}.$$

The first equation gives $\Delta\nu = -1 - d_1$, and the last gives $d_2 = 1 - d_1$. Substitution into the second yields

$$e(1 - d_1) - 1 - d_1 = -e \quad \Rightarrow \quad d_1 = \frac{2e - 1}{e + 1} = 1.193175736.$$

Therefore $d_2 = -0.193175736$, $\Delta\nu = -2.193175736$. The primal components sum to one, cancelling the original infeasibility under a full step.

2. Test the nonlinear residual, not just the linear model.

At the full trial,

$$x_1 = (1.193175736, 0.806824264), \qquad \nu_1 = -2.193175736,$$

the actual residual norm is 1.105386522, below the Armijo residual limit $0.75(3.064156670) = 2.298117503$. Accept the step. The new objective is approximately 5.538317, greater than the infeasible initial value 3.718282.

3. Rebuild the KKT system and update both variables.

k	x_1	x_2	r_p	ν_k	f	$\|r\|_2$
0	0	1	-1	0	3.718282	3.064157
1	1.193176	0.806824	0	-2.193176	5.538317	1.105387
2	1.002368	0.997632	0	-2.66834	5.436579	0.071223
3	1	1	0	-2.718274	5.436564	1.077×10^{-5}
4	1	1	0	-2.718282	5.436564	0

Full steps are accepted throughout these iterations. The first step removes affine infeasibility, but not the nonlinear dual residual. Later steps bring x to $(1, 1)$ and ν to $-e$. After iteration 3, the primal point is already extremely accurate, but the residual is still about 1.1×10^{-5}, mainly because the multiplier has not yet converged. Another step is needed for a tighter primal-dual stopping tolerance.

Why the Steps Work

The Jacobian contains the objective Hessian because the nonlinear part of the KKT residual is ∇f. The affine primal equation is corrected exactly by a full step, while the dual equation is only locally linearized. Updating the multiplier with the same step length is essential to reducing stationarity error. The last table explains why observing accurate coordinates or small objective changes alone is not a reliable residual-based stopping rule.

Appendix B

Revision Reference and Reproducible Algorithms

B.1 Method Selection at a Glance

Situation	Suitable approach	Essential check
Independent simple bounds	Parametrization or a bound-aware solver	Does the map attain required endpoints?
Easy equality elimination	Reduced-variable optimization	All branches and reduced domains retained
Small symbolic constrained problem	Lagrange or KKT equations	Regularity, all active bounds, signs, classification
Smooth equality problem	Augmented Lagrangian or equality Newton	Feasibility and stationarity residuals
Strict interior available, convex inequalities	Log barrier with Newton centering	Strict domain and duality-gap control
Affine equalities, feasible initial point	Feasible-start Newton	Positive tangent curvature and $Ad = 0$
Affine equalities, infeasible initial point	Infeasible-start Newton	Primal-dual residual reduction

B.2 Core Equations

KKT and curvature

$$L = f + \nu^T h + \mu^T g, \qquad \mu \geq 0, \quad h = 0, \quad g \leq 0, \quad \mu_j g_j = 0,$$
$$\nabla_x L = 0, \qquad H_L = \nabla^2 f + \sum_i \nu_i \nabla^2 h_i + \sum_j \mu_j \nabla^2 g_j.$$

For equality constraints, test H_L on $\ker J_h$. For inequalities, critical directions require the appropriate active-set and multiplier conditions; treating every active inequality as an equality in all directions can over-restrict the test.

Penalty, augmented Lagrangian, and barrier

$$P_Q = \|h\|^2 + \|g_+\|^2,$$
$$\mathcal{L}_\rho = f + \nu^T h + \frac{\rho}{2}\|h\|^2, \qquad \nu_+ = \nu + \rho h(x_+),$$
$$F_t = tf - \sum_j \log(-g_j), \qquad \mu_j = -\frac{1}{tg_j}, \qquad \mu_j g_j = -\frac{1}{t}.$$

For an exactly centered convex log-barrier problem, $f(x_t) - p^* \leq m/t$.

Newton systems and residuals

$$K = \begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix}, \quad r = \begin{bmatrix} \nabla f + A^T \nu \\ Ax - b \end{bmatrix}.$$

$$K \begin{bmatrix} d \\ \widehat{\nu}_+ \end{bmatrix} = \begin{bmatrix} -\nabla f \\ 0 \end{bmatrix} \quad \text{at a feasible point,} \qquad K \begin{bmatrix} \Delta x \\ \Delta \nu \end{bmatrix} = -r \quad \text{in the increment formulation.}$$

B.3 A Practical Infeasible-Start Newton Implementation

The following educational routine expects `oracle(x)` to return the objective value, gradient, and Hessian. It assumes affine equalities, a suitable twice-differentiable objective, and nonsingular KKT systems. The objective routine should reject points outside its domain. The routine backtracks on residual norm and updates the primal and dual variables together.

```
import numpy as np

def equality_newton(oracle, A, b, x0, nu0=None,
                    tol=1e-10, maxiter=50, c=0.25, beta=0.5):
    A = np.asarray(A, dtype=float)
    b = np.asarray(b, dtype=float)
    x = np.asarray(x0, dtype=float).copy()
    nu = np.zeros(A.shape[0]) if nu0 is None else np.array(nu0, float)
    trace = []
    for k in range(maxiter):
        value, gradient, hessian = oracle(x)
        rp = A @ x - b
        rd = gradient + A.T @ nu
        residual = np.r_[rd, rp]
        norm0 = np.linalg.norm(residual)
        trace.append((k, x.copy(), nu.copy(), value, norm0))
        if max(np.linalg.norm(rp), np.linalg.norm(rd)) <= tol:
            return x, nu, trace
        K = np.block([[hessian, A.T],
                      [A, np.zeros((A.shape[0], A.shape[0]))]])
        step = np.linalg.solve(K, -residual)
        dx, dnu = step[:x.size], step[x.size:]
        alpha = 1.0
        for trial in range(60):
            xt, nut = x + alpha * dx, nu + alpha * dnu
            try:
                ft, gt, _ = oracle(xt)
                rt = np.r_[gt + A.T @ nut, A @ xt - b]
                accept = (np.isfinite(ft) and np.all(np.isfinite(rt))
                          and np.linalg.norm(rt) <= (1-c*alpha)*norm0)
            except (ValueError, FloatingPointError, OverflowError):
                accept = False
            if accept:
                x, nu = xt, nut
                break
            alpha *= beta
        else:
            raise RuntimeError("Residual backtracking failed")
    raise RuntimeError("Maximum iterations reached")

def exponential_oracle(x):
    e = np.exp(x)
    return e.sum(), e, np.diag(e)

x, nu, trace = equality_newton(
```

```
    exponential_oracle, [[1., 1.]], [2.], [0., 1.])
assert np.allclose(x, [1., 1.], atol=1e-9)
assert np.allclose(nu, [-np.e], atol=1e-9)
```

For larger problems, use structured or sparse KKT factorizations, scaling, and appropriate regularization. Explicit inversion is unnecessary. The illustrative implementation is intentionally limited to affine equalities and does not implement a complete general-purpose inequality solver.

B.4 Verification Checklist for Worked Calculations

1. **Model:** distinguish capacities from mandatory usage; retain nonnegativity and domain restrictions.
2. **Candidates:** solve every necessary branch, including zero-coordinate and endpoint cases.
3. **Feasibility:** substitute candidate points into every original equality and inequality.
4. **Stationarity and signs:** use the stated Lagrangian convention and check $\mu \geq 0$.
5. **Optimality:** use convex sufficiency, tangent curvature, a dual certificate, or a justified global comparison.
6. **Numerics:** compute residuals from the unrounded iterates and monitor subproblem convergence.
7. **Interpretation:** report original variables, units, active constraints, and any local nature of sensitivity estimates.

B.5 Context and Further Reading

The mathematical scope is based on the following supplied course documents. The explanations, examples, calculations, and figures needed to use this volume are contained in the volume itself.

Context document	Topic coverage
`EN3330_L07.pdf`, pp. 3–41	Feasible sets, transformations, variable elimination, Lagrange multipliers, carton and production examples
`EN3330_L08.pdf`, pp. 3–87	Regularity, KKT conditions, active sets, duality, penalties, augmented Lagrangian, log barriers, central paths
`EN3330_L09.pdf`, pp. 3–43	Newton models and decrement, damping, feasible and infeasible-start equality Newton, economic dispatch

The reference page numbers above count physical PDF pages. Repeated presentations of the same equation are consolidated. Globality claims are justified in the worked solutions rather than inferred from stationarity alone.

Further theoretical references:

- Mykel J. Kochenderfer and Tim A. Wheeler, *Algorithms for Optimization*, MIT Press, 2019.
- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.
- Andreas Antoniou and Wu-Sheng Lu, *Practical Optimization: Algorithms and Engineering Applications*, Springer, 2007.